

CS 3151 – Software Engineering

Requirements Engineering Report

Phase 1 – Midterm Submission

Tempo – AI Marketing Co-Pilot

Riman Almarri , Mays Almousa , Ghaliah Badokhon
S24109297 · S24109301 · S24109427

Instructor: **Dr. Passent Elkafrawy**

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Abstract

Tempo is an AI-powered, bilingual (Arabic/English) marketing co-pilot for solo marketers, SMEs, freelance content creators and boutique agencies in Saudi Arabia. It brings Saudi-dialect AI content creation, scheduling to Snapchat, X and Instagram, unified social and website analytics (Google Search Console) and automated bilingual reporting into one dashboard, priced in Saudi Riyals. This report is the final version of the project's analysis phase (Requirements Engineering, Phase 1). It presents the problem to be solved, the market study and feasibility study, the business model canvas, the user personas, the elicitation methods used and the complete set of requirements, the use cases, the context diagram and the project management plan. All lab and SRS submissions were revised for this version: every requirement was checked against the SMART criteria and rewritten where it was not Specific, Measurable, Achievable, Relevant and Time-bound (Section 8.3).

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1 Introduction

1.1 Project Overview

Develop Tempo, an AI-powered marketing co-pilot that helps solo marketers, SMEs, freelance content creators and agencies in Saudi Arabia manage social media and website marketing from one bilingual (Arabic/English) dashboard. Tempo replaces a fragmented daily routine — logging into Snapchat, X, Instagram and Google Search Console separately, rewriting English-first AI output into Saudi Arabic, and building reports by hand — with a single workflow: plan and draft, schedule, measure, report.

Features include Saudi-dialect AI content generation (Najdi and Hijazi as well as Modern Standard Arabic and English), a unified content calendar with scheduled publishing, a unified social and website analytics dashboard, automated weekly bilingual reports with branded PDF export, Saudi-native timing and trend insights, client workspaces and role-based team access for agencies, and SAR-denominated pricing with a freemium entry point.

1.2 Problem to Be Solved

Saudi Arabia has more than 25 million active social media users and a fast-growing e-commerce sector driven by Vision 2030, yet the tools its marketers rely on were built for Western markets. The market study (survey of 47 Saudi marketing professionals in Riyadh and Jeddah, Q3 2026) identified three gaps:

- **Weak Arabic AI:** existing AI writers produce generic or formal Arabic that must be rewritten into Saudi dialect, costing about 4 hours per week.
- **Fragmented platforms:** no tool unifies Snapchat, X, Instagram and Google Search Console in a single Arabic-language dashboard, so data is consolidated manually (6 hours per week for a solo marketer and 12+ hours per week for an agency team).
- **Unaffordable, USD-priced tools:** enterprise suites cost more than SAR 900 per month and foreign-currency pricing creates unpredictable costs for SMEs.

Problem statement. Saudi marketers and small agencies lose between 4 and 12 hours every week rewriting AI output into Saudi dialect, switching between four platforms and building reports by hand, because no affordable, SAR-priced tool combines Saudi-dialect content, scheduling for the Saudi platform mix and unified analytics.

1.3 Objectives

The project objectives are written in SMART form (each has a measurable target and a date):

- Generate three ready-to-edit Saudi-dialect (Najdi, Hijazi), MSA or English drafts from a short brief in under 30 seconds, in the MVP released by February 2027.
- Schedule and publish content to X and Instagram from one unified calendar by the public launch (February 2027), and to Snapchat by April 2027 (subject to platform approval).
- Combine social metrics and Google Search Console website data in one bilingual dashboard by February 2027.
- Reduce weekly reporting time for a solo marketer from about 6 hours to under 1 hour through automated bilingual reports, measured with closed-beta users (January–February 2027).
- Support agencies with client workspaces, role-based access control and branded PDF reports by June 2027.

- Offer SAR pricing (Starter at SAR 189/month, Agency at SAR 499/month) with a free trial from the public launch, and reach 150 paying accounts by the end of 2027.
- Comply with the Saudi Personal Data Protection Law (PDPL) and host all customer data in a Saudi cloud region from day one of the closed beta.

1.4 Scope

In scope (three-year plan): a mobile-responsive web application (progressive web app) for solo marketers, SMEs, freelance creators and boutique agencies in Riyadh, Jeddah and Dammam, followed by a GCC pilot. **Out of scope:** native mobile apps for the MVP, large in-house enterprise marketing departments (identified as a user group but deferred), paid-advertising management, and training a proprietary language model.

2 Sustainability Goals

2.1 SDGs Alignment

- **SDG 8 (Decent Work and Economic Growth):** Raises the productivity of SMEs and freelancers by saving roughly 4–11 hours of manual work per week, letting small businesses grow without hiring a specialist for every channel.
- **SDG 9 (Industry, Innovation and Infrastructure):** Brings affordable AI and analytics technology to small Saudi businesses and builds local Arabic-language AI capability.
- **SDG 5 (Gender Equality):** Many Saudi home-based and online businesses are led by women who sell through Instagram and Snapchat; low-cost, Arabic-first tools strengthen their digital and economic participation.
- **SDG 10 (Reduced Inequalities):** SAR pricing and a free tier narrow the gap between large enterprises with expensive marketing suites and small businesses with limited budgets.

2.2 Vision 2030 Contribution

Tempo aligns with Saudi Arabia's Vision 2030 through:

- **SME Growth:** Supporting the Vision 2030 goal of increasing the SME contribution to GDP by giving small businesses enterprise-grade marketing intelligence at an SME price.
- **Digital Transformation:** Moving marketing work from spreadsheets and manual consolidation to an automated, data-driven platform.
- **E-commerce and Digital Economy:** Helping online retailers reach Saudi consumers on the platforms where they discover brands (Snapchat, X and Instagram).
- **Arabic Content and Local Identity:** Producing culturally relevant content in Saudi dialects and around local occasions (Ramadan, Eid, Saudi National Day, Founding Day).
- **Women and Youth Empowerment:** Enabling women-led businesses, freelance creators and young entrepreneurs to compete professionally online.
- **Local Tech Talent and Data Sovereignty:** Building a Saudi software team and hosting customer data inside the Kingdom in line with national data-protection requirements.

3 Market Study

3.1 Saudi Digital Marketing Environment

Saudi Arabia is home to more than 25 million active social media users and a rapidly expanding e-commerce sector driven by Vision 2030. Snapchat, X and Instagram dominate how Saudi consumers discover brands, but the available tools were designed elsewhere: they lack Arabic-language AI, do not integrate the Saudi-dominant platforms with website analytics, and are priced in foreign currency.

3.2 Market Size

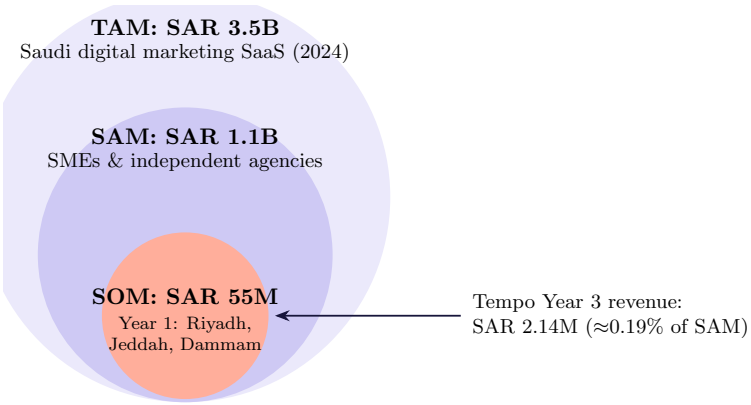


Figure 1: Market Size (TAM / SAM / SOM)

The Saudi digital marketing SaaS market is worth SAR 3.5 billion (TAM, 2024), with a serviceable addressable market of SAR 1.1 billion (SMEs and independent agencies) and a Year 1 obtainable market of SAR 55 million. The SOM is the market study's estimate of what is reachable, not a sales forecast; Tempo's own revenue plan is deliberately much smaller (Year 3 revenue of SAR 2.14M is about 0.19% of the SAM).

3.3 Target Customer Segments

Segment	Profile	Key pain point	Use
Solo marketers & SME owners	Manage own social and website marketing (persona: Nora)	Four separate platforms; English-first AI; USD pricing	Daily
Boutique agencies	Manage multiple client accounts (persona: Khalid)	12+ hours/week compiling data; no affordable multi-client workspaces; formal Arabic instead of client dialect	Daily
Freelance content creators	Produce Arabic content for several brands (persona: Lama)	Slow drafting and rewriting; one calendar per brand	Several times a week

Table 1: Target Segments (initial focus: Riyadh, Jeddah and Dammam, then GCC)

3.4 Competitor Analysis

Four tools dominate the digital marketing SaaS space globally but fall short of Saudi requirements. **Hootsuite** (SAR 370/month) and **Buffer** (SAR 56/month) offer social scheduling and basic analytics but provide no Arabic AI content generation and lack Snapchat API integration. **Sprout Social** (SAR 934/month) offers enterprise reporting and bilingual posting but targets

large organisations and prices out Saudi SMEs. **Jasper AI** (SAR 184/month) generates Arabic marketing copy but is a pure content tool with no scheduling, analytics or website performance features.

Existing solutions fall into three categories: (1) global scheduling tools (Hootsuite, Buffer, Later) that lack Arabic AI and Snapchat support; (2) enterprise social suites (Sprout Social, Brandwatch) priced above SAR 900/month; and (3) AI writing assistants (Jasper, Copy.ai) that generate Arabic text but offer no publishing or analytics. None combine bilingual content generation, Saudi platform scheduling and website performance tracking in a single, affordable product.

Feature	Tempo	Hootsuite	Buffer	Sprout	Jasper
Monthly price (SAR)	189	370	56	934	184
Entry plan at or below SAR 200	✓	✗	✓	✗	✓
Bilingual Arabic/English content	✓	✗	✗	✓	✓
Saudi-dialect AI content generation	✓	✗	✗	✗	✗
Content scheduling & calendar	✓	✓	✓	✓	✗
Snapchat integration	✓	✗	✗	–	✗
Automated reporting & analytics	✓	○	○	✓	✗
Unified social + website (Search Console) analytics	✓	✗	✗	✗	✗
Saudi-native market insights	✓	✗	✗	✗	✗
SAR pricing (no FX risk)	✓	✗	✗	✗	✗

Table 2: Competitive Feature Comparison (✓ = yes, ✗ = no, ○ = basic only, – = not confirmed)

3.5 Customer Feedback Analysis

Reviews of competing tools were analysed across the Apple App Store, Google Play Store and G2, alongside a survey of 47 Saudi marketing professionals in Riyadh and Jeddah (Q3 2026).

Positive feedback (what users value).

- All-in-one dashboards that eliminate switching between tools — cited by 68% of respondents
- AI that drafts social captions in Arabic without manual translation — 61%
- Affordable pricing in SAR that avoids foreign-exchange risk — 54%
- Automated weekly performance reports that save an average of 6 hours per week

Negative feedback and market gaps.

- Existing tools lack Arabic AI that captures Saudi colloquial dialect and cultural nuance, forcing marketers to rewrite AI output (about 4 hours per week).
- No tool integrates Snapchat, X Analytics, Instagram Insights and Google Search Console in one Arabic-language dashboard; marketers log into four platforms and consolidate data by hand.

The survey sample is small and limited to two cities, so it supports the existence of demand but not its exact size; the closed beta will test willingness to pay.

3.6 Market Opportunity and Differentiation

Tempo differentiates itself by:

- Offering the only bilingual Arabic/English AI content suite with Saudi-dialect focus and direct Snapchat, X and Instagram scheduling.

- Unifying social media and website analytics in one dashboard — replacing four logins with one automated weekly report in Arabic and English.
- Pricing entirely in SAR, with a Starter plan at SAR 189/month and a freemium trial.

These advantages can be copied over time, so speed to market and local content quality are the defensible parts. Tempo is sold directly online with self-service sign-up, an in-product trial and local payment methods.

3.7 Marketing Strategy

Positioning: "the Saudi marketer's co-pilot" — the only affordable tool that writes in Saudi Arabic, publishes to Snapchat, X and Instagram, and reports on social and website results together. Messaging stresses time saved rather than technology. **Pricing:** a free tier lowers risk for SMEs; Starter at SAR 189 is close to Jasper AI (content only) and far below Hootsuite and Sprout Social. About one in nine free users is assumed to convert to paying.

Phase	Timing	Focus	Main actions
1. Pre-launch	Q4 2026	Validate, build waitlist	Landing page, prototype demos in Riyadh and Jeddah, beta invitations
2. Saudi launch	Q1 2027	Solo marketers, SMEs, creators	Public freemium and Starter release, Arabic social content, creator partnerships
3. Agency expansion	H1 2027	Boutique agencies	Agency workspaces and branded reports, agency outreach, case studies
4. GCC expansion	H2 2027 onward	Selected GCC markets	Pilot in one or two GCC countries after Gate 3; localise dialects and payments

Table 3: Go-to-Market Phases

Channels. Arabic-first content on Snapchat, Instagram and X (using Tempo itself); free workshops with chambers of commerce, coworking spaces and e-commerce communities; direct agency outreach via LinkedIn and referrals; university entrepreneurship programmes and incubators; bilingual SEO; and a referral programme (one free month per referred paying account).

Budget and retention. Year 1 marketing spend is SAR 90,000 (content and paid social 35,000; creator partnerships 15,000; events 15,000; referral incentives 15,000; website and SEO 10,000), rising to SAR 180,000 and SAR 300,000. Retention relies on habits the product creates — the Monday weekly report, the content calendar and agency workspaces. The team tracks activation (first post scheduled within 7 days), free-to-paid conversion, monthly churn and hours saved.

4 Feasibility Study

This section assesses Tempo from product, market, technical, operational and economic perspectives and recommends a phased, MVP-first launch with controlled, stage-gated spending.

4.1 Executive Summary

Tempo is **feasible on conditions, not unconditionally**. It is technically and operationally achievable by a small team using proven technologies and a commercial LLM API, and the market gap is well supported. The financial case depends on reaching about **614 paying accounts**, which makes controlled, stage-gated spending essential. Pre-launch costs are about SAR 115,000 and total funding of SAR 1.0 million over two rounds covers the plan. Revenue is projected at SAR 176,625, 865,950 and 2,142,660 in Years 1 to 3, with net results of SAR (329,091), (328,949) and

153,050. Monthly operating break-even is reached in mid-2029; full recovery of invested capital falls after the three-year horizon.

4.2 Description of Products and Services

Tempo is a subscription-based, mobile-responsive web platform that acts as a marketing co-pilot for the Saudi market.

Feature	What it does	Main users
Bilingual AI content	Drafts captions, post copy and campaign ideas from a short brief, in Arabic or English, with editable output.	All segments
Saudi-localised AI content	Tunes output to Saudi colloquial dialect (Najdi, Hijazi), local occasions and cultural references instead of formal MSA.	Solo marketers, SMEs, agencies
Content calendar & scheduling	One calendar for Snapchat, X and Instagram with scheduled publishing (e.g. the 8–10 PM peak window).	All segments
Unified analytics dashboard	Reach, engagement and follower data from the three platforms alongside website performance in one Arabic/English view.	All segments
Website analytics	Google Search Console data (queries, clicks, impressions, pages) next to social results.	SMEs, agencies
Automated reporting	Weekly reports generated automatically in Arabic and English; branded PDF export with client logo.	Solo marketers, agencies
Saudi-native insights	Trend and timing insights drawn from the Saudi platform mix.	All segments
Agency client workspaces	Separate workspace per client with fast switching and no repeated log-in.	Agencies
Role-based team access	Owner, admin, editor and viewer roles; drafts can be assigned and approved inside the tool.	Agencies, SMEs with teams
Mobile-responsive dashboard	Full dashboard usable on phones for approvals on the move; no native app for the MVP.	All segments

Table 4: Core Features of Tempo

Value for each user group. *Solo marketers* (persona: Nora) gain Saudi-dialect drafts in about 30 seconds and a weekly report without manual compilation, cutting reporting from 6 hours to under 1. *SMEs* get an affordable SAR-priced tool covering content, scheduling and website results. *Agencies* (persona: Khalid, 8 clients) manage every client from one dashboard and issue branded bilingual reports in minutes, saving about 11 hours per week. *Freelance creators* (persona: Lama) produce Arabic content faster and keep one calendar across several brands.

Plan	Price (SAR/mo)	Intended for	Main limits and inclusions
Free / trial	0	Evaluation, creators	One workspace, limited AI generations and scheduled posts; no PDF branding
Starter	189	Solo marketers, SMEs	Three social platforms, Search Console, AI content, scheduling, weekly bilingual report
Agency (proposed)	499	Boutique agencies	Multiple client workspaces, team roles, branded bilingual PDF reports, higher AI allowance

Table 5: Proposed Plan Structure (prices exclude 15% VAT)

4.3 Market Feasibility

- **Market value:** Saudi digital marketing SaaS market of SAR 3.5 billion (TAM, 2024), with a serviceable addressable market of SAR 1.1 billion (SMEs and independent agencies) and a Year 1 obtainable market of SAR 55 million.
- **Demand evidence:** 68% of surveyed professionals value all-in-one dashboards, 61% value AI that drafts Arabic captions without manual translation, and 54% value SAR pricing.
- **Unique selling points:** the only bilingual Arabic/English AI content suite with Saudi-dialect focus; direct Snapchat, X and Instagram scheduling plus Google Search Console in one dashboard; one automated weekly bilingual report replacing four platform logins; SAR pricing from SAR 189/month with a freemium entry.
- **Conservative targets:** Year 3 revenue (SAR 2.14M) is only about 0.19% of the SAM. The detailed market analysis is given in Section 3.

4.4 Technical Feasibility

Tempo is a software-only product, so technology is both the main cost and the main source of risk. The approach is to build the product-specific parts (workspaces, content studio, dashboard, reports) and buy everything else (cloud infrastructure, AI models, payments, email). A single modular back end (not microservices) keeps the system simple for a small team: a web front end calls a REST API, a background worker handles publishing, data collection and report generation, and an AI service layer isolates the LLM provider from the rest of the product.

Layer	Proposed technology	Purpose and notes
Front end	React/Next.js with TypeScript, Tailwind CSS, PWA features	RTL Arabic and LTR English from one codebase; responsive layout
Back end / API	Node.js with TypeScript (NestJS), REST API, BullMQ job queue on Redis	Business logic, workspaces, scheduling, queued publishing and data sync
Database	PostgreSQL; Redis cache/queues; S3-compatible object storage	Workspace ID on every record for tenant separation; media and PDFs
AI / LLM	Commercial LLM API behind a provider-independent interface; Saudi-dialect prompt templates	Drafts and insight summaries; human editing before publishing; provider can be swapped
Analytics	Scheduled collectors, daily metrics per account; Recharts/Chart.js; SQL materialised views	Unified dashboard and report data without a separate analytics platform
Auth	Email/password + optional MFA, OAuth 2.0 for connected accounts, RBAC	Secure sign-in, per-workspace permissions
Reports	Server-side HTML-to-PDF (headless Chromium) with Arabic fonts and logo templates	Bilingual, branded PDF reports with correct RTL layout
Hosting	Containers on a cloud provider with a Saudi region (e.g. Google Cloud Dammam, Oracle Cloud Jeddah/Riyadh)	Managed DB, backups, CI/CD; Saudi data residency

Table 6: Proposed Technology Stack

Integration	Use in Tempo	Constraint and planned response
Instagram (Meta Graph API)	Publishing, insights	Requires Business/Creator accounts and Meta app review. Apply in Sprint 0; support professional accounts only.
X API	Posting, analytics	Tiered, rate-limited and paid; terms have changed before. Budget API fees, cache data, queue requests.
Snapchat APIs	Scheduling, analytics	More restricted, approval-based access. Release after X and Instagram; CSV import fallback for analytics.
Google Search Console API	Website performance	Read-only OAuth with quotas. Low risk; daily data pulls.
LLM provider API	Content, summaries	Per-use cost, rate limits and dialect quality vary. Prompt tests, usage caps per plan, provider-independent layer.

Table 7: Third-Party Integrations and Associated Risks

- **AI output quality:** generative models can produce off-tone text, and Saudi dialects (especially Hijazi) are less covered than MSA. Tempo treats AI output as a draft: users edit before scheduling, a Saudi-dialect evaluation set is scored by native speakers before each release, and user ratings feed prompt improvements.
- **Security:** access tokens encrypted at rest, minimum permission scopes, HTTPS everywhere, hashed passwords, rate limiting, audit logs, dependency scanning and an external penetration test before launch.
- **Privacy:** design follows the Saudi PDPL (consent, purpose limitation, deletion on request); legal counsel confirms obligations before launch.
- **Scalability:** a stateless API, queue-based workers and a managed database scale horizontally. The main constraints are third-party rate limits and AI cost per user, not compute.

4.5 Operational Feasibility

Tempo starts as a three-person founding team with part-time marketing support and grows to eight people by Year 3. Roles are combined at first so that each hire is added only when workload and revenue justify it.

Role	Main responsibilities	Y1	Y2	Y3
Product / Project Manager (founder)	Backlog, sprint planning, requirements, partners and funding	1	1	1
Backend & AI-integration Dev. (founder)	API, database, platform integrations, publishing queue, security	1	1	1
Frontend & UX Developer (founder)	Bilingual RTL/LTR interface, dashboard, calendar, workspaces	1	1	1
AI / ML Engineer	Prompt design, dialect evaluation, AI cost control, content safety	–	1	1
Marketing & Sales Executive	Launch campaigns, SME and agency outreach, onboarding	0.5	1	1
Customer Support Specialist	Arabic/English chat and email support, help content	–	1	1
UI/UX Designer	Design system, usability testing, report templates	Contract	–	1
Agency & GCC Sales Executive	Agency accounts, GCC pilots, partner channels	–	–	1
Total full-time equivalents		**3.5**	**6**	**8**

Table 8: Roles and Responsibilities

The founders can deliver the MVP; Year 1 depends on their time commitment and on keeping scope tight. The main staffing risks — founder overload before the Year 2 hires and dependence on a few key people — are reduced through documentation, code review and shared ownership of each integration. Accounting, legal and penetration testing are outsourced.

4.6 Economic Feasibility

All figures are in Saudi Riyals and are **estimates based on assumptions**. Only the Starter price, the market sizes and the survey results come from the market study; every other number is an academic assumption to be replaced with beta data. Year 1 is 2027, Year 2 is 2028 and Year 3 is 2029. Prices exclude VAT.

4.6.1 Key Assumptions

Item	Assumption
Plans and mix	Starter SAR 189 and Agency SAR 499 per month, monthly billing. Starter share: 85% (Y1), 80% (Y2), 75% (Y3)
ARPU	Y1: $0.85 \times 189 + 0.15 \times 499 =$ SAR 235.50; Y2: SAR 251.00; Y3: SAR 266.50 per month
Paying accounts (year end)	Y1: 150; Y2: 450; Y3: 950 (quarterly path in Table 11)
Free / trial users	Eight per paying account on average (about 11% conversion)
AI usage cost	SAR 16 per paying account and SAR 1.5 per free user per month (SAR 28 per paying account incl. its free users)
Payment processing	2.5% of revenue
Cloud hosting	SAR 24,000 (Y1), 54,000 (Y2), 108,000 (Y3)
Depreciation	Equipment over 3 years, straight line
Zakat	Simplified to 2.5% of positive operating profit
Accounts payable	One month of AI, hosting and payment costs at year end; no receivables (card payment in advance)

Table 9: Planning Assumptions

4.6.2 Start-up Costs

Item (pre-launch, Oct – Dec 2026)	SAR
Founder stipends ($3 \times 6{,}000 \times 3$ months)	54,000
UI/UX design contract (prototype, design system)	15,000
Company registration, trademark and legal setup	8,000
Development tools, cloud sandbox and domains	6,000
AI API testing credits (prompt and dialect tuning)	4,000
Beta and waitlist marketing	10,000
Laptops and equipment (capitalised, $3 \times 6{,}000$)	18,000
Total start-up costs	**115,000**

Table 10: Start-up Cost Breakdown (SAR 18,000 capitalised, SAR 97,000 expensed)

4.6.3 Revenue Projection

	Year 1 (2027)	Year 2 (2028)	Year 3 (2029)
Paying accounts, end Q1 / Q2 / Q3 / Q4	20 / 55 / 100 / 150	210 / 280 / 360 / 450	540 / 650 / 790 / 950
Account-months	750	3,450	8,040
ARPU (SAR per month)	235.50	251.00	266.50
Subscription revenue (SAR)	**176,625**	**865,950**	**2,142,660**

Table 11: Paying Accounts and Subscription Revenue

Example, Year 1: account-months $= \frac{0+20}{2} \cdot 3 + \frac{20+55}{2} \cdot 3 + \frac{55+100}{2} \cdot 3 + \frac{100+150}{2} \cdot 3 = 750$; revenue $= 750 \times 235.50 =$ SAR 176,625.

4.6.4 Income Statement

SAR	Year 1	Year 2	Year 3
Sales (subscription revenue)	176,625	865,950	2,142,660
Cost of sales (AI usage, payment fees, hosting)	(49,416)	(172,249)	(386,686)
Gross profit	**127,209**	**693,701**	**1,755,974**
Salaries	(276,000)	(684,000)	(1,056,000)
Marketing	(90,000)	(180,000)	(300,000)
Software and tools	(18,000)	(30,000)	(48,000)
Legal, accounting and compliance	(24,000)	(30,000)	(40,000)
Coworking and office	(24,000)	(48,000)	(72,000)
Security testing and audits	(10,000)	(25,000)	(40,000)
Miscellaneous (5% of non-salary expenses)	(8,300)	(15,650)	(25,000)
Depreciation	(6,000)	(10,000)	(18,000)
Administrative and marketing expenses	**(456,300)**	**(1,022,650)**	**(1,599,000)**
Operating profit	**(329,091)**	**(328,949)**	**156,974**
Zakat	0	0	(3,924)
Net profit	**(329,091)**	**(328,949)**	**153,050**

Table 12: Estimated Income Statement (brackets are deductions or losses)

Gross margin is about 72%, 80% and 82% in Years 1 to 3. Losses in Years 1 and 2 reflect team and marketing spend ahead of revenue; Year 3 shows a profit of SAR 153,050 (Figure 2).

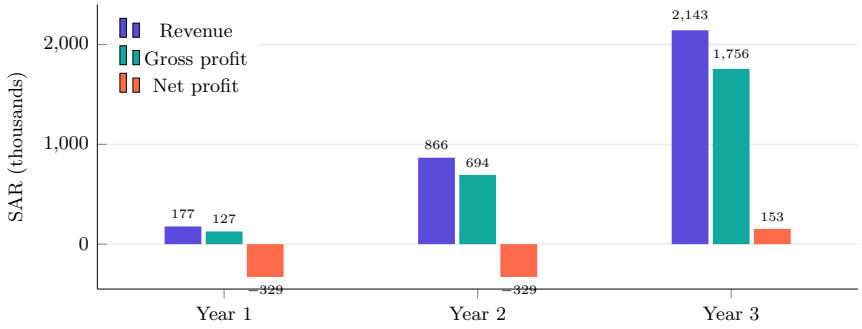


Figure 2: Three-Year Financial Projection

4.6.5 Balance Sheet and Financing

SAR (year end)	Year 1	Year 2	Year 3
Current assets (cash)	166,027	245,314	414,158
Fixed assets (net)	12,000	14,000	20,000
Total assets	**178,027**	**259,314**	**434,158**
Current liabilities (payables and zakat)	4,118	14,354	36,148
Long-term liabilities	0	0	0
Equity: paid-in capital	600,000	1,000,000	1,000,000
Equity: retained earnings (deficit)	(426,091)	(755,040)	(601,990)
Total equity	**173,909**	**244,960**	**398,010**
Total liabilities and equity	**178,027**	**259,314**	**434,158**

Table 13: Estimated Balance Sheet

The plan is financed entirely by equity: SAR 600,000 in Q4 2026 (assumed SAR 100,000 from the founders and SAR 500,000 from a seed investor, grant or incubator programme) and a further SAR 400,000 in January 2028. No source of funding has been secured yet; these are planning assumptions. The opening retained deficit of SAR 97,000 is the pre-launch expense.

4.6.6 Cash Flow Statement

SAR	Year 1	Year 2	Year 3
Cash flows from operating activities	(318,973)	(308,713)	192,844
Cash flows from investing activities (equipment)	0	(12,000)	(24,000)
Cash flows from financing activities	0	400,000	0
Net cash flows	**(318,973)**	**79,287**	**168,844**
Cash balance at the beginning of the period	485,000	166,027	245,314
Cash balance at the end of the period	**166,027**	**245,314**	**414,158**

Table 14: Estimated Cash Flow Statement

Opening cash on 1 January 2027 is SAR 485,000 (SAR 600,000 raised less SAR 115,000 pre-launch spending). Cash falls to SAR 166,027 at the end of Year 1, which is why the second funding round is needed before Year 2.

Role (SAR per month, fully loaded)	**Year 1**	**Year 2**	**Year 3**
Product / Project Manager (founder)	6,000	9,000	12,000
Backend & AI-integration Developer (founder)	6,000	9,000	12,000
Frontend & UX Developer (founder)	6,000	9,000	12,000
AI / ML Engineer	–	14,000	15,000
Marketing & Sales Executive	4,000	9,000	9,500
Customer Support Specialist	–	7,000	7,500
UI/UX Designer	1,000	–	11,000
Agency & GCC Sales Executive	–	–	9,000
Total monthly	**23,000**	**57,000**	**88,000**
Total annual	**276,000**	**684,000**	**1,056,000**

Table 15: Monthly Staff Cost Assumption (Year 1 marketing is 0.5 FTE; designer on a SAR 12,000 annual retainer)

4.6.7 Break-even and Sensitivity

SAR	**Year 1**	**Year 2**	**Year 3**
Fixed monthly costs (admin. expenses + hosting) ÷ 12	40,025	89,721	142,250
Variable cost per paying account-month	33.89	34.27	34.66
Contribution per account-month (ARPU − variable cost)	201.61	216.72	231.84
Paying accounts needed to break even	**199**	**414**	**614**
Average paying accounts in the year	62	288	670

Table 16: Operating Break-even Analysis

On Year 3 costs Tempo needs about 614 paying accounts. The plan passes that level during Q2 of Year 3 (650 accounts at the end of Q2), so **monthly operating break-even is expected in mid-2029**. Cumulative results are still negative at the end of Year 3 (retained deficit SAR 601,990), so investors recover their capital only after the three-year horizon. Contribution per account is high (about SAR 230), so the model is sound if customer numbers are reached.

Sensitivity. If paying accounts reach only 70% of plan, revenue falls to SAR 123,637, 606,165 and 1,499,862 and results to SAR (374,454), (553,259) and (402,218) before zakat. Cash would then run out around the end of Year 2 unless costs are reduced, so hiring must be tied to actual customer numbers. In a downside case the team can delay the AI/ML, support and Year 3 hires, reducing salary costs by about SAR 252,000 in Year 2 and SAR 510,000 in Year 3.

4.7 System Requirements

4.7.1 Hardware Requirements

Component	Specification
Development Workstation	CPU: Intel Core i7 / Apple M2 or equivalent (8 cores); RAM: 16 GB minimum (32 GB recommended for containers); Storage: 512 GB SSD; Display: Full HD; test phones (Android and iPhone) for RTL/responsive checks
Production (Cloud, Saudi region)	Application: 2–4 containers, 2 vCPU / 4 GB RAM each, auto-scaling; Workers: 2 containers, 2 vCPU / 4 GB (publishing, sync, PDF rendering); Database: managed PostgreSQL, 4 vCPU / 16 GB, 200 GB SSD, daily backups, standby replica; Cache/queue: managed Redis 2 GB; object storage 500 GB for media and PDFs
Staging & Testing	Scaled-down copy of production (1 app, 1 worker, small DB); sandbox/test accounts on Instagram, X, Snapchat and Search Console
End User (Client)	Any modern laptop/desktop or smartphone (screen width $\geq$ 360 px); stable internet connection (4G or better)

Table 17: Hardware Specifications

4.7.2 Software Requirements

Category	Requirements
Operating Systems	Development: Windows 11, macOS 14+ or Ubuntu 22.04+; Production: Linux containers (Ubuntu 22.04 LTS base images)
Development Tools	Node.js 20 LTS, TypeScript 5, Next.js 14+, NestJS 10+, Tailwind CSS; PostgreSQL 16, Redis 7, BullMQ, Prisma ORM; Docker, GitHub Actions (CI/CD), VS Code
AI & Reporting	Commercial LLM API (via provider-independent interface); headless Chromium (Puppeteer) for PDF; Arabic web fonts (e.g. IBM Plex Sans Arabic); Recharts / Chart.js for charts
External APIs	Meta Graph API (Instagram), X API v2, Snap APIs, Google Search Console API; local payment gateway supporting mada, Visa/Mastercard and Apple Pay
Testing & Security	Jest (unit), Playwright (end-to-end), k6 (load testing); OWASP ZAP, dependency scanning (npm audit / Dependabot)
Design & Management	Figma (UI/UX), Jira or GitHub Projects (Scrum board), Google Workspace; client side: latest Chrome, Safari, Edge or Firefox

Table 18: Software Specifications

4.8 Risk Analysis

Risk	L	I	Mitigation
Dependence on Snapchat, X, Instagram APIs (access denied, delayed or removed)	H	H	Start approvals in Sprint 0; launch X and Instagram first; adapter per platform; CSV fallback
Platform policy, pricing and rate-limit changes	H	M	Monitor announcements; cache data; budget API fees
Competition: global tools add Arabic AI or cut prices	M	H	Focus on dialect quality, local insights, agency workflow; SAR pricing; release quickly
Slow acquisition / low free-to-paid conversion	M	H	Stage-gated spending; early agency pilots; weekly conversion tracking
AI output quality or unsuitable content	M	H	Human review; native-speaker evaluation set; content filters; ratings
Weak Saudi/Hijazi dialect quality	M	M	Dialect-specific prompts; city-based user tests; AI/ML engineer owns quality
Data privacy non-compliance (PDPL)	L	H	Privacy by design; minimal data; legal review; Saudi hosting
Cybersecurity incident or token theft	L	H	Encryption, least privilege, MFA, audit logs, penetration test
AI usage costs exceed assumptions	M	M	Usage caps per plan; caching; cheaper models for simple tasks
Scalability and performance	L	M	Stateless design, queued jobs, load tests before launch
Schedule delay or founder overload	M	M	Tight MVP scope; two-week sprints; cut scope, not date
Funding not secured	M	H	Spend only against gates; cash reserve; grants and incubators

Table 19: Risk Register (L = likelihood, I = impact; H/M/L = high/medium/low)

4.9 Findings and Recommendations

We recommend that the project **proceed as an MVP-first, stage-gated launch**. Tempo is not a low-risk project; its financial return depends on reaching roughly 600 paying accounts.

Dimension	Conclusion	Feasible if
Technical	Feasible	Snapchat access is confirmed or replaced by a fallback, and AI dialect quality passes beta evaluation.
Operational	Feasible with limits	Three founders commit full time; hires follow revenue and gates.
Market	Feasible	Beta users confirm survey results and willingness to pay SAR 189; agencies adopt the Agency plan.
Financial	Conditionally feasible	About SAR 1.0M is raised in two stages; accounts grow to roughly 600+ within three years.

Table 20: Feasibility Conclusions

Gate	When	Evidence required	Decision if met
1. Public launch	Feb 2027	∼40 beta users; Arabic output usable with light editing; X and Instagram publishing stable; security test passed	Open paid access, start Starter marketing
2. Agency & hiring	Jun 2027	≥55 paying accounts; monthly churn ≤5%; ≥5 agencies in pilot	Release agency features; plan Year 2 hires
3. GCC pilot	Sep 2027	≥100 paying accounts; positive contribution; second round progressing	Start GCC pilot in one country
4. Scale	Jan 2028	∼150 paying accounts; second round closed	Hire AI/ML engineer, support and sales

Table 21: Stage Gates for Releasing Spending

If a gate is missed, the team extends the previous phase, changes pricing or positioning, or reduces costs before spending more. This limits the amount at risk to about SAR 115,000 before launch and about SAR 440,000 by the end of Year 1.

5 Business Model Canvas

The Business Model Canvas summarises how Tempo creates, delivers and captures value. Every block is taken from the market study (Section 3) and the feasibility study (Section 4).

Key Partners
- Meta (Instagram Graph API), X, Snap and Google (Search Console API)
- Commercial LLM provider
- Cloud host with a Saudi region (e.g. Google Cloud Dammam)
- Local payment gateway (mada, Visa/Mastercard, Apple Pay)
- Chambers of commerce, coworking spaces, incubators and university programmes
- Creator partners and referral users

Key Activities
- Saudi-dialect prompt design and native-speaker evaluation
- Building and maintaining platform integrations
- Product development in two-week sprints
- Onboarding, support and agency outreach

Key Resources
- Bilingual founding team (product, back end, front end)
- Saudi-dialect prompt library and evaluation set
- Approved platform API access
- Saudi-region cloud infrastructure
- Brand and customer data

Value Propositions
- Three Saudi-dialect (Najdi, Hijazi), MSA or English drafts in under 30 s
- One calendar to publish to Snapchat, X and Instagram
- Social + website (Search Console) analytics in one Arabic/English dashboard
- Automated weekly bilingual report: 6 h → under 1 h
- Client workspaces and branded PDFs for agencies
- SAR pricing from SAR 189 with a free tier
- Data hosted in KSA (PDPL)

Customer Relationships
- Self-service sign-up with free trial
- In-app onboarding (first post in 10 min)
- Arabic/English chat and email support
- Account management for agencies
- Habit features: Monday report, calendar

Channels
- Website and in-product trial
- Arabic content on Snapchat, Instagram and X
- Workshops with chambers of commerce and coworking spaces
- LinkedIn agency outreach and referrals
- Bilingual SEO; referral programme

Customer Segments
- Solo marketers and SME owners (Nora)
- Boutique digital agencies (Khalid)
- Freelance content creators (Lama)

First: Riyadh, Jeddah, Dammam; then a GCC pilot.

Deferred: enterprise marketing departments.

Cost Structure
- Salaries: SAR 276k (Y1) → 1.06M (Y3)
- Marketing: SAR 90k (Y1) → 300k (Y3)
- AI usage: SAR 16 per paying account per month
- Cloud hosting: SAR 24k (Y1) → 108k (Y3); platform API fees
- Legal, compliance and security testing
- Start-up costs: SAR 115k

Revenue Streams
- Monthly subscriptions in SAR (excl. 15% VAT): Starter SAR 189 · Agency SAR 499
- Freemium: about 1 in 9 free users converts
- Revenue: SAR 176,625 (Y1), 865,950 (Y2), 2,142,660 (Y3)
- Break-even at about 614 paying accounts (mid-2029)

Figure 3: Tempo Business Model Canvas

6 User Research and Personas

Following a user-centred design approach, we identified Tempo's user groups, built realistic personas and mapped each persona to required features.

6.1 User Groups

User Group	Role	Frequency	Technical Level
Solo Marketers & SME Owners	Manage own social media and website marketing on Saudi platforms	Daily	Intermediate
Digital Marketing Agencies	Manage multiple client social media accounts	Daily	Intermediate–Advanced
Freelance Content Creators	Produce Arabic content for brands on Snapchat and Instagram	Several times/week	Beginner–Intermediate
Enterprise Marketing Depts.	In-house teams at large corporations needing unified analytics (deferred beyond Year 3)	Daily	Advanced

Table 22: Potential User Groups

6.2 Personas

6.2.1 Persona 1: Nora Al-Shamrani (Solo Marketer)

Persona Name	Nora Al-Shamrani
Age	28
Education	Bachelor of Business Administration – Marketing, King Abdulaziz University, Jeddah
Role	Digital Marketing Manager at a mid-sized Saudi e-commerce fashion brand
Goals	Manage Snapchat, X and Instagram campaigns from one bilingual dashboard; reduce weekly reporting time from 6 hours to under 1 hour; generate Arabic AI content that resonates with Saudi audiences.
Motivations	Impress her manager with clear, data-driven ROI reports; stay competitive as Saudi consumers discover brands on Snapchat and Instagram Reels; avoid expensive tools priced in USD.
Pain Points	Switches between 4 platforms (Snapchat, X Analytics, Instagram Insights, Google Search Console); AI tools produce generic English copy she must translate and rewrite in Saudi dialect; USD subscriptions create unpredictable costs.
Technical Skill Level	Intermediate – comfortable with SaaS marketing tools; not a developer; prefers an intuitive UI over technical setup.
Usage Context	Daily desktop use in her Riyadh office and mobile use during commutes; posts mainly between 8–10 PM Saudi time (peak Snapchat engagement); needs mobile approvals.

Table 23: Persona: Nora Al-Shamrani

6.2.2 Persona 2: Khalid Al-Otaibi (Agency Account Manager)

Persona Name	**Khalid Al-Otaibi**
Age	35
Education	Master of Business Administration (MBA), King Fahd University of Petroleum & Minerals
Role	Senior Account Manager at a boutique digital marketing agency in Jeddah (manages 8 client accounts)
Goals	Manage all client accounts from one dashboard; generate branded bilingual reports weekly; scale team capacity by automating content calendars; win new SME clients with AI-powered analytics.
Motivations	Differentiate his agency in a competitive Jeddah market; reduce junior staff hours on manual reporting; deliver measurable ROI to retail and F&B clients who demand Arabic outputs.
Pain Points	No tool supports per-client workspaces at an agency-accessible price; compiling data from four platforms takes his team 12+ hours per week; Arabic AI tools produce formal MSA, not the Hijazi dialect his clients prefer.
Technical Skill Level	Advanced – experienced with many SaaS platforms and API integrations; expects team access controls and role permissions.
Usage Context	Mainly desktop in the Jeddah office; generates client reports every Monday morning; uses dashboards during client calls; his team of 4 needs simultaneous access to different client workspaces.

Table 24: Persona: Khalid Al-Otaibi

6.2.3 Persona 3: Lama Al-Harbi (Freelance Content Creator)

Persona Name	**Lama Al-Harbi**
Age	24
Education	Bachelor of Graphic Design, Imam Abdulrahman Bin Faisal University, Dammam
Role	Freelance content creator producing Snapchat and Instagram content for five small Saudi brands (cafés, beauty, home-based businesses)
Goals	Deliver more client content in less time; keep one calendar across all her brands; send clients simple performance summaries to justify her fees.
Motivations	Grow her freelance income and client list; build a professional reputation; keep tool costs low while she is starting out.
Pain Points	Writes every caption from scratch for five different brand voices; juggles separate spreadsheets and phone reminders for each brand's schedule; clients ask for numbers she has to screenshot from each app.
Technical Skill Level	Beginner–Intermediate – confident on mobile apps and design tools; little experience with analytics or integrations.
Usage Context	Mostly on her phone between shoots and meetings; plans content on Sunday evenings on a laptop; starts on the Free plan and upgrades when she has more clients.

Table 25: Persona: Lama Al-Harbi

6.3 Persona-to-Requirements Mapping

Persona	Feature Needed	Priority	Req. ID
Nora (Solo Marketer)	Bilingual Arabic/English AI content drafting	High	RQ-F3
Nora (Solo Marketer)	One-click scheduling to Snapchat, X & Instagram	High	RQ-F4, F5
Nora (Solo Marketer)	Automated weekly performance report in Arabic	High	RQ-F8
Nora (Solo Marketer)	SAR-denominated pricing with freemium tier	High	RQ-F12
Nora (Solo Marketer)	Mobile-responsive dashboard	Medium	RQ-NF6
Khalid (Agency Manager)	Multi-account client workspace management	High	RQ-F9
Khalid (Agency Manager)	Branded bilingual PDF report export	High	RQ-F8
Khalid (Agency Manager)	Unified social + website analytics (Search Console)	High	RQ-F6
Khalid (Agency Manager)	Team role-based access control	Medium	RQ-F10
Khalid (Agency Manager)	Saudi-native platform insights (Snapchat, X trends)	High	RQ-F7
Lama (Freelance Creator)	Multiple brand voices / dialect selection in AI drafts	High	RQ-F3
Lama (Freelance Creator)	One calendar across several brands	Medium	RQ-F4, F9
Lama (Freelance Creator)	Free tier and simple upgrade path	High	RQ-F12

Table 26: Persona-to-Feature Mapping

6.4 Usage Scenarios

Nora's Monday planning session. Nora opens Tempo in her Riyadh office and types a brief for this week's Eid al-Adha campaign in the AI Content Studio. The AI generates three Snapchat captions and two Instagram Reels descriptions in Saudi Arabic within 30 seconds. She edits one caption to add a local Riyadh reference, then schedules all posts across Snapchat, X and Instagram for 9 PM Saudi time. She clicks "Generate Weekly Report" and Tempo compiles last week's reach, engagement and website traffic into a bilingual PDF, which she shares with her manager in two clicks.

Khalid's weekly client reporting. Every Monday Khalid opens his agency dashboard and switches between eight client workspaces. For three high-priority retail clients he clicks "Generate Branded Report" and Tempo produces bilingual PDF reports with each client's logo. He assigns the week's content drafts to two junior team members through role-based workspaces, eliminating email chains — saving 11 hours per week.

Lama's Sunday content sprint. On Sunday evening Lama opens Tempo on her laptop, switches between her five brand workspaces and generates a week of captions for each in the right tone (playful Hijazi for a Jeddah café, elegant MSA for a perfume brand). She drags the drafts into each brand's calendar at the suggested evening slots, then on Thursday sends each client a one-page summary straight from her phone.

7 Requirements Elicitation Methods

Requirements were gathered from users, from the market and from technical sources, then validated through reviews. Table 27 lists each method, what the team did and which requirements it

produced.

Method	What we did	Key findings	Requirements produced
Questionnaire (survey)	Online survey of 47 Saudi marketing professionals in Riyadh and Jeddah, Q3 2026	68% want one dashboard; 61% want Arabic AI captions; 54% want SAR pricing	RQ-F3, RQ-F6, RQ-F12
Analysis of user reviews	Reviews of Hootsuite, Buffer, Sprout Social and Jasper AI on the App Store, Google Play and G2	No Saudi dialect; no Snapchat; four logins; manual reports (about 6 h/week)	RQ-F3, RQ-F4, RQ-F5, RQ-F8
Competitor (product) analysis	Feature-by-feature comparison of five tools (Table 2)	No tool unifies Saudi platforms with Search Console or prices in SAR	RQ-F6, RQ-F7, RQ-F12
Personas and scenarios	Three personas (Nora, Khalid, Lama) and usage scenarios built from the survey and review data (Section 6)	Different needs per segment: speed, multi-client work, low cost	RQ-F9, RQ-F10, RQ-NF6
Use cases and user stories	Twelve user stories and six detailed use cases written from the personas (Section 9)	Main flows, alternative flows and actors	RQ-F1–F13; L4-FR-A01–D05
Document analysis	Study of the Meta, X, Snap and Google Search Console API documentation, the Saudi PDPL and VAT rules	API approvals, rate limits, data residency, VAT invoices	RQ-F2, RQ-F12, RQ-NF3, RQ-NF4, RQ-NF5
Team brainstorming and prioritisation	Team workshops to group requirements into modules and rank them (MoSCoW: Must / Should Have)	MVP scope versus later phases	Priorities in Sections 9 and 10
Prototyping (planned validation)	Clickable bilingual prototype shown to potential users in Riyadh and Jeddah (Nov – Dec 2026) and closed beta with about 40 users (Jan – Feb 2027)	To confirm dialect quality, usability and willingness to pay	Validates RQ-F3, RQ-NF6

Table 27: Elicitation Methods and Their Output

Validation. Every requirement was reviewed against the SMART criteria (Specific, Measurable, Achievable, Relevant, Time-bound) and traced to a persona and a use case (Tables 32 and 49). Requirements that failed the review were rewritten; the changes are listed in Section 8.3. The main limitation is the survey sample (47 people in two cities), so the closed beta is used to confirm the findings.

8 Requirements Specification

All requirements below are the final, SMART version. Unless a different date is given, each must be delivered and tested by the public launch (Gate 1, February 2027); agency requirements (RQ-F9, RQ-F10 and branding in RQ-F8) are due by June 2027 (Gate 2). Rows shaded green were rewritten in this revision.

8.1 Functional Requirements

ID	Description
RQ-F1	Users can register with email and password, verify the email through a link valid for 24 hours, log in, reset a forgotten password through a link valid for 30 minutes, and enable multi-factor authentication (MFA) with an authenticator app.
RQ-F2	Owners and Admins can connect and revoke Instagram, X and Snapchat accounts (OAuth 2.0) and one Google Search Console property per workspace; when a connection is revoked, its tokens are deleted within 1 minute.
RQ-F3	From a brief of up to 500 characters, users can generate exactly three editable content drafts within 30 seconds, choosing language and dialect (Najdi, Hijazi, MSA, English), platform, tone and occasion; users can rate each draft from 1 to 5 stars.
RQ-F4	Users can create, edit, reschedule (drag and drop) and delete posts in a unified content calendar with day, week and month views; for the selected week the calendar highlights the three time slots per platform with the highest average engagement in the last 30 days.
RQ-F5	The system automatically publishes scheduled posts to the selected platforms at the scheduled time and records the result (published / failed) with a timestamp and the platform's post ID.
RQ-F6	The system displays a unified dashboard of social metrics (reach, engagement rate, followers) and website metrics (clicks, impressions, queries, average position), refreshed at least once every 24 hours and filterable by platform and by date range (last 7 days, last 30 days, or a custom range of up to 12 months).
RQ-F7	Every day, the system shows Saudi-native insights calculated from the workspace's last 30 days of data: the three best posting times per platform (Saudi time), the five Search Console queries with the largest increase in clicks compared with the previous 30 days, and the Saudi occasions (Ramadan, Eid al-Fitr, Eid al-Adha, Founding Day, National Day) in the next 30 days.
RQ-F8	Every Monday at 09:00 (Saudi time) the system generates the previous week's performance report in Arabic, English or both within 5 minutes and makes it available as a PDF; Agency-plan users can apply client branding (logo and two brand colours).
RQ-F9	Agency-plan users can create at least 10 client workspaces and switch between them from the top bar in under 2 seconds without logging out.
RQ-F10	The system enforces role-based access control with four roles (Owner, Admin, Editor, Viewer) per workspace on every request; when the optional approval step is on, a post created by an Editor cannot be scheduled until an Owner or Admin approves it.
RQ-F11	The system sends in-app and email notifications within 1 minute of these events: post published or failed, report ready, approval requested, and a connected account expiring within the next 7 days.
RQ-F12	Users can subscribe to, upgrade, downgrade or cancel the Free, Starter (SAR 189/month) and Agency (SAR 499/month) plans and pay in SAR by mada, Visa/Mastercard or Apple Pay; a VAT invoice (15%) is issued within 1 minute of each successful payment.
RQ-F13	Users can switch the interface between Arabic (right-to-left) and English (left-to-right) with one click from any screen; the change applies to all screens without logging out and is saved to the user's profile.

Table 28: Functional Requirements (final SMART version)

8.2 Non-Functional Requirements

ID	Category	Description
RQ-NF1	Performance	Dashboard pages load within 2 seconds and AI drafts are returned within 30 seconds (95th percentile), measured with a k6 load test of 1,000 concurrent users before the public launch.
RQ-NF2	Availability	The system maintains 99.5% monthly uptime (at most about 3.6 hours of downtime per month), excluding maintenance announced 48 hours in advance and never scheduled between 6 PM and 12 AM Saudi time; measured by an external uptime monitor from the public launch.
RQ-NF3	Reliability	99% of scheduled posts are published within 2 minutes of their scheduled time; failed API calls are retried up to 3 times with back-off. Measured monthly from the publishing log.
RQ-NF4	Security	All traffic uses TLS 1.2+; access tokens are encrypted at rest (AES-256); passwords are hashed (bcrypt/Argon2); every create, edit, publish and delete action is audit-logged; an external penetration test runs before launch and every year, and all Critical and High findings are fixed before launch.
RQ-NF5	Privacy	The system complies with the Saudi PDPL, stores data in a Saudi cloud region and deletes user data within 30 days of a request.
RQ-NF6	Usability & Localisation	Full RTL Arabic and LTR English support on 100% of screens; responsive from 360 px screen width; WCAG 2.1 AA contrast; at least 80% of closed-beta testers schedule a first post within 10 minutes of sign-up.
RQ-NF7	Scalability	The system supports 1,000 concurrent users and 10,000 connected social accounts without architectural change, verified by load test before Gate 4 (January 2028).
RQ-NF8	Maintainability	AI providers and social platforms sit behind interchangeable interfaces so that a new LLM provider can be added in at most 5 working days without changes outside the AI module; core modules keep at least 70% unit-test coverage, checked automatically in CI/CD on every merge.

Table 29: Non-Functional Requirements (final SMART version)

8.3 SMART Revision Log

As required for the midterm, all lab and SRS submissions were revised. Table 30 lists every requirement that was not SMART, the problem found and the fix (S = Specific, M = Measurable, A = Achievable, R = Relevant, T = Time-bound). It also records the conflicts that were found between the Requirements Specification and the Lab 4 SRS, which now use the same targets.

ID	Problem in previous version	Fix in final version
RQ-F1	"Reset their password" had no limits (M)	Email link valid 24 h; reset link valid 30 min; MFA via authenticator app
RQ-F2	"Revoke" did not say what happens to the tokens (S, M)	Tokens deleted within 1 minute; one Search Console property per workspace
RQ-F3	"At least three" drafts with no time limit (M)	Exactly three drafts within 30 s; brief up to 500 characters; rating 1–5 stars
RQ-F4	"Suggested peak-time slots" was not defined (S, M)	Three highest-engagement slots per platform from the last 30 days

ID	Problem in previous version	Fix in final version
RQ-F5	Result recorded but no evidence defined (M)	Timestamp and platform post ID stored for each result
RQ-F6	No refresh rate; "filterable by date" unlimited (M)	Refreshed every 24 h; 7 days, 30 days or custom up to 12 months
RQ-F7	"Trending local topics" could not be tested (S, M)	Three best times per platform, top five rising queries, occasions in next 30 days, refreshed daily
RQ-F8	"Weekly, automatically" had no day, time or duration (T, M)	Monday 09:00 Saudi time, ready within 5 minutes; branding = logo + two colours
RQ-F9	"Multiple" workspaces; no switching time (M)	At least 10 workspaces; switch in under 2 s
RQ-F10	Approval step did not say what it blocks (S)	Editor's post cannot be scheduled until an Owner or Admin approves
RQ-F11	No delivery time; "expiring" not defined (M)	Sent within 1 minute; expiring = within the next 7 days
RQ-F12	Payment methods and invoice timing missing (S, M)	mada, Visa/Mastercard, Apple Pay; 15% VAT invoice within 1 minute
RQ-F13	"At any time" could not be tested (M)	One click from any screen, applies to all screens, saved to profile
RQ-NF1	No test conditions (M, A)	Measured with k6 at 1,000 concurrent users before launch
RQ-NF2	"Announced maintenance" unlimited (M)	48 h notice; never during 6 PM – 12 AM peak; external monitor
RQ-NF4	"Actions are audit-logged" too vague; test with no pass rule (S, M)	Listed logged actions; Critical and High findings fixed before launch
RQ-NF6	"A new user" — no sample or pass rate (M)	At least 80% of closed-beta testers within 10 minutes
RQ-NF7	"Year 3" with no test (T, M)	Load test before Gate 4 (January 2028)
RQ-NF8	"Interchangeable" could not be measured (M)	New provider in at most 5 working days; coverage checked on every merge
L4-FR-B01	"Up to 3" variants conflicted with "at least three" in RQ-F3 (S)	Exactly three variants within 30 s
L4-FR-B02, B03	"Top-performing posts" and "engagement-potential score" not defined (S, M)	Top 10 posts by engagement rate; score from 0 to 100
L4-FR-B06, C03–C05, D01	No timing for alerts, sync, top-post card or PDF (T)	Times added (2 min, 06:00, 07:00, 1 h, 60 s)
L4-NFR-01	3 s target conflicted with RQ-NF1 (2 s)	Aligned to 2 s at the 95th percentile
L4-NFR-02	10 s target conflicted with RQ-NF1, RQ-F3 and UC-01 (30 s)	Aligned to 30 s at the 95th percentile, as in the project objectives
L4-NFR-03	Uptime only during peak hours, conflicting with RQ-NF2	Aligned to 99.5% monthly uptime with no planned maintenance at peak
L4-NFR-07, 08	"Fully support" and "new users" not measurable (M)	100% of screens checked each release; 80% of beta testers within 10 min

ID	Problem in previous version	Fix in final version
L4-NFR-10	5,000 concurrent users conflicted with RQ-NF7 (1,000) and was not achievable for the planned team (A)	Aligned to 1,000 concurrent users, tested before Gate 4
L4-NFR-11	"Without changes to other modules" had no measure (M)	New provider in at most 5 working days

Table 30: SMART Revision Log

9 Use Case Analysis

9.1 Actors

- **Primary:** Marketer (solo marketer, SME owner or freelance creator), Agency Manager (workspace Owner/Admin), Team Member (Editor or Viewer, including a client given view-only access)
- **Secondary:** LLM Provider (AI API), Social Platforms (Instagram, X, Snapchat APIs), Google Search Console, Payment Gateway, Scheduler (system timer that triggers publishing, data sync and weekly reports)

9.2 Use Case Titles

ID	Use case title	Primary actor	Priority	Implementation phase
UC-01	Generate AI Content	Marketer / Team Editor	Critical	Phase 1 – MVP (Nov 2026 – Feb 2027)
UC-02	Schedule & Publish Posts (extended by *Approve Post*)	Marketer / Team Editor	Critical	Phase 1 – MVP (Snapchat: Feb – Apr 2027)
UC-03	View Unified Analytics	Marketer / Agency Manager / Viewer	High	Phase 1 – MVP (Dec 2026 – Feb 2027)
UC-04	Generate Bilingual Report	Marketer / Agency Manager	High	Phase 1 (weekly report); Phase 3 (branding)
UC-05	Manage Client Workspaces & Team Roles	Agency Manager	Medium	Phase 3 – Agency (Apr – Jun 2027)
UC-06	Manage Account, Security & Subscription	User	Critical	Phase 1 – MVP (Nov 2026 – Feb 2027)

Table 31: Use Case Titles, Priority and Implementation Timeline

The Lab 4 SRS (Section 10) consolidates these into four simplified use cases: L4-UC01 Connect Account & Configure Workspace, L4-UC02 Create & Schedule Content with AI, L4-UC03 Monitor Unified Analytics, and L4-UC04 Generate & Export PDF Report.

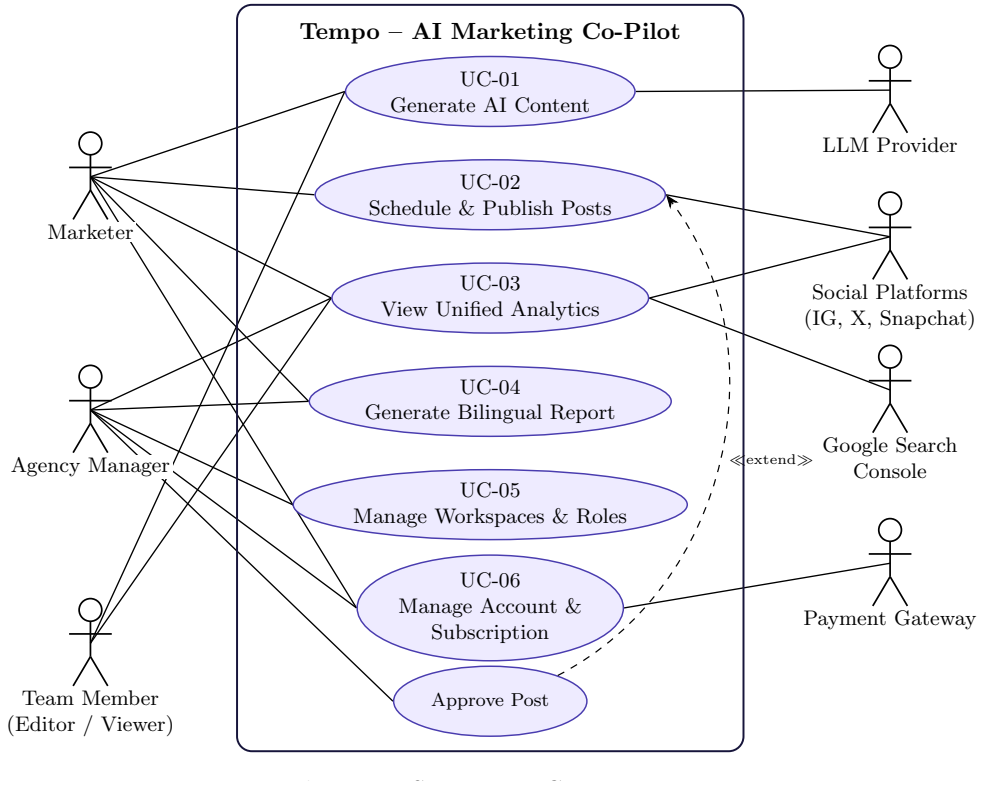


Figure 4: System Use Case Diagram

9.3 User Stories

- As a solo marketer, I want to generate Snapchat and Instagram captions in Saudi Arabic dialect so that I can connect authentically with my audience without hours of manual translation and rewriting.
- As a solo marketer, I want to view all my social media and website analytics in a single Arabic-language dashboard so that I can make data-driven decisions without logging into four platforms.
- As a solo marketer, I want to schedule one campaign to Snapchat, X and Instagram at 9 PM from one calendar so that I reach my audience at peak time without posting manually.
- As a solo marketer, I want a weekly report generated automatically in Arabic so that I can share results with my manager in two clicks.
- As an agency account manager, I want to manage all my client accounts from a single dashboard so that I can switch between client workspaces without logging out and back in.
- As an agency account manager, I want to generate branded bilingual PDF reports automatically so that I can deliver client-ready insights in under 5 minutes instead of 3 hours.
- As an agency account manager, I want to assign drafts to junior team members and approve them before publishing so that client content is always checked.
- As an agency account manager, I want to give a client view-only access so that they can see results without being able to change anything.
- As a freelance creator, I want to keep separate brand voices and calendars for each client so that I can manage five brands from one account.
- As a freelance creator, I want to start on a free plan and upgrade in SAR when I get more clients so that tool costs grow with my income.

- As a team editor, I want to be notified when my draft is approved or rejected so that I can act quickly.
- As an account owner, I want to enable multi-factor authentication and revoke connected accounts so that my brand's social accounts stay secure.

9.4 Traceability Matrix

	UC-01 AI Content	**UC-02** Schedule	**UC-03** Analytics	**UC-04** Report	**UC-05** Workspaces	**UC-06** Account
RQ-F1						X
RQ-F2		X	X			X
RQ-F3	X					
RQ-F4		X				
RQ-F5		X				
RQ-F6			X			
RQ-F7	X		X			
RQ-F8				X		
RQ-F9					X	
RQ-F10		X			X	
RQ-F11		X		X	X	
RQ-F12						X
RQ-F13	X		X	X		X
RQ-NF1	X		X	X		
RQ-NF2		X	X			
RQ-NF3		X				
RQ-NF4					X	X
RQ-NF5			X			X
RQ-NF6	X		X	X		
RQ-NF7		X	X			
RQ-NF8	X	X				

Table 32: Traceability Matrix for Tempo Use Cases

9.5 Detailed Use Cases

9.5.1 UC-01: Generate AI Content

This use case describes how a user turns a short brief into ready-to-edit, Saudi-dialect content drafts.

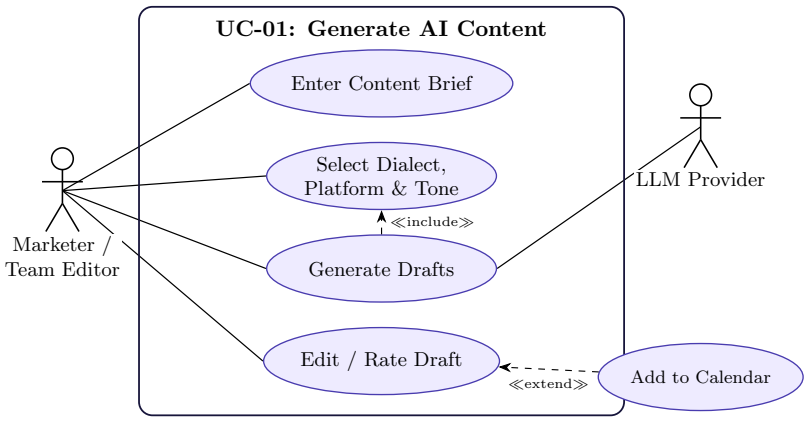


Figure 5: Use Case Diagram: Generate AI Content

Use Case	**UC-01 Generate AI Content**
Actors	Marketer or Team Editor (primary); LLM Provider (secondary)
Preconditions	User is logged in, has a workspace and remaining AI credits for the plan.
Trigger	User clicks “Generate” in the AI Content Studio.
Main Flow	1. User opens the AI Content Studio in the active workspace. 2. User enters a brief (up to 500 characters) and selects dialect (Najdi, Hijazi, MSA, English), platforms, tone and occasion. 3. System validates the brief and checks the AI credit balance. 4. System builds a dialect-specific prompt and sends it to the LLM Provider. 5. LLM Provider returns generated text; system applies content filters and platform length limits. 6. System displays three drafts within 30 seconds, with character counts and hashtag/timing suggestions. 7. User edits, rates (1–5 stars) or regenerates a draft. 8. User saves the draft or adds it to the calendar (extends to UC-02).
Alternative Flows	3a. Credits exhausted: system shows upgrade options (UC-06). 4a. Provider does not answer within 30 s: system retries once with the fallback provider, otherwise shows an error and keeps the brief. 5a. Filter flags unsuitable content: the draft is withheld and regenerated.
Postconditions	Brief and drafts are stored in the workspace; AI usage is recorded.
Requirements	RQ-F3, RQ-F7, RQ-F13, RQ-NF1, RQ-NF6, RQ-NF8

Table 33: Use Case Description: UC-01 Generate AI Content

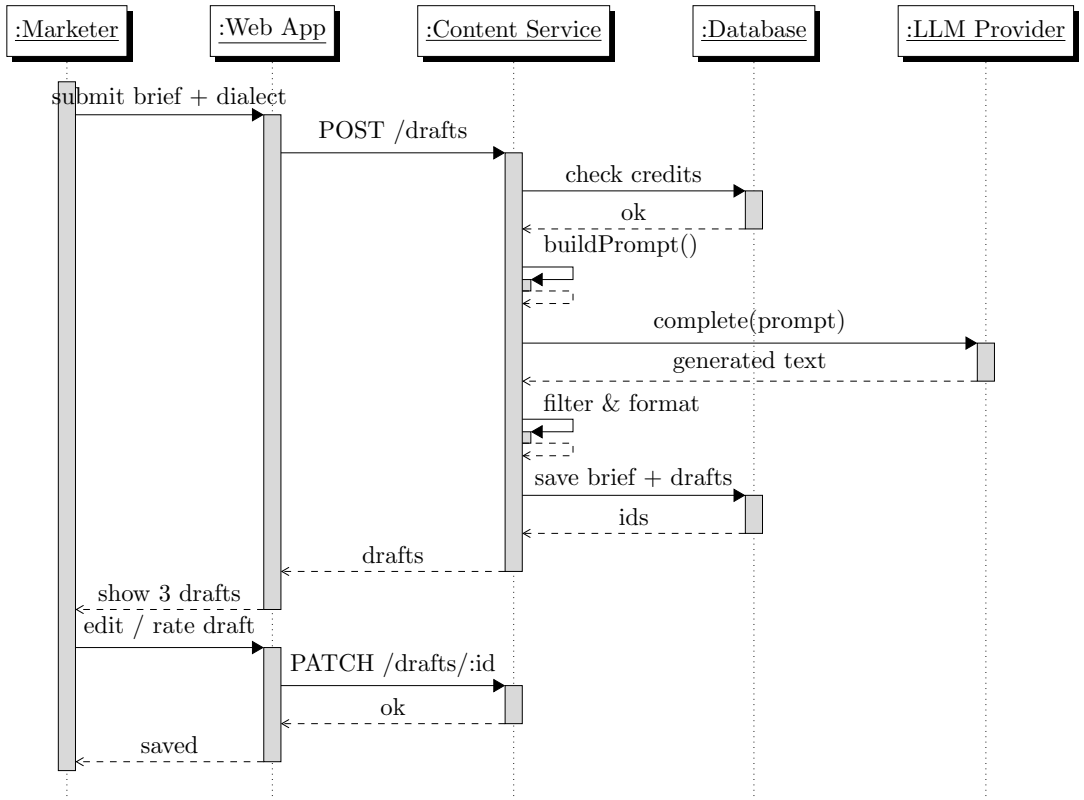


Figure 6: Sequence Diagram for Generate AI Content

9.5.2 UC-02: Schedule & Publish Posts

This use case covers scheduling a draft to one or more platforms and its automatic publication at the chosen time.

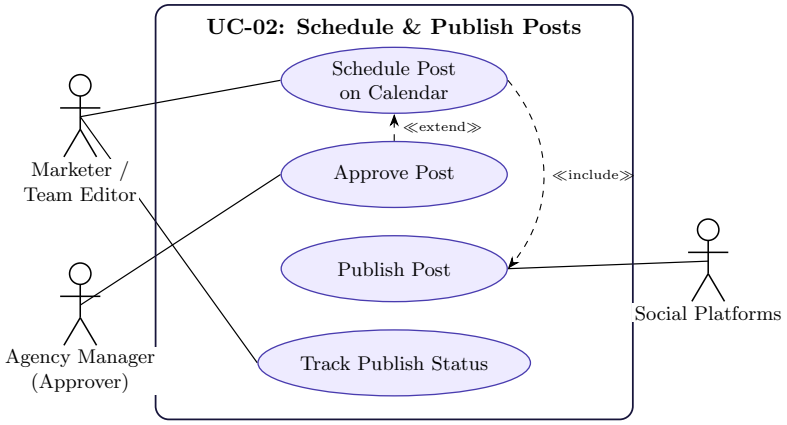


Figure 7: Use Case Diagram: Schedule & Publish Posts

Use Case	UC-02 Schedule & Publish Posts
Actors	Marketer / Team Editor (primary); Agency Manager as approver; Social Platforms and Scheduler (secondary)
Preconditions	A draft exists; at least one social account is connected with a valid token.
Trigger	User drags a draft onto the calendar or clicks “Schedule”.
Main Flow	1. User selects platforms and a date/time; system highlights the three best slots of the week (usually 8–10 PM). 2. System validates connected accounts, media format and caption length per platform. 3. If the workspace requires approval, system notifies the approver, who approves the post. 4. System saves the post as scheduled and adds a job to the publishing queue. 5. At the scheduled time the Scheduler triggers the worker. 6. Worker publishes through the platform’s API and receives the external post ID. 7. System marks the post published and notifies the user, starts metric collection and writes the audit log.
Alternative Flows	2a. Token expired: system asks the user to reconnect the account. 3a. Approver rejects: draft returns to the editor with comments. 6a. API error or rate limit: worker retries 3 times with back-off, then marks failed and alerts the user within 2 minutes. 6b. Snapchat not yet approved: system offers “export and post manually”.
Postconditions	Post is live on the platform(s) and its status is recorded; metric collection is scheduled.
Requirements	RQ-F2, RQ-F4, RQ-F5, RQ-F10, RQ-F11, RQ-NF2, RQ-NF3, RQ-NF7, RQ-NF8

Table 34: Use Case Description: UC-02 Schedule & Publish Posts

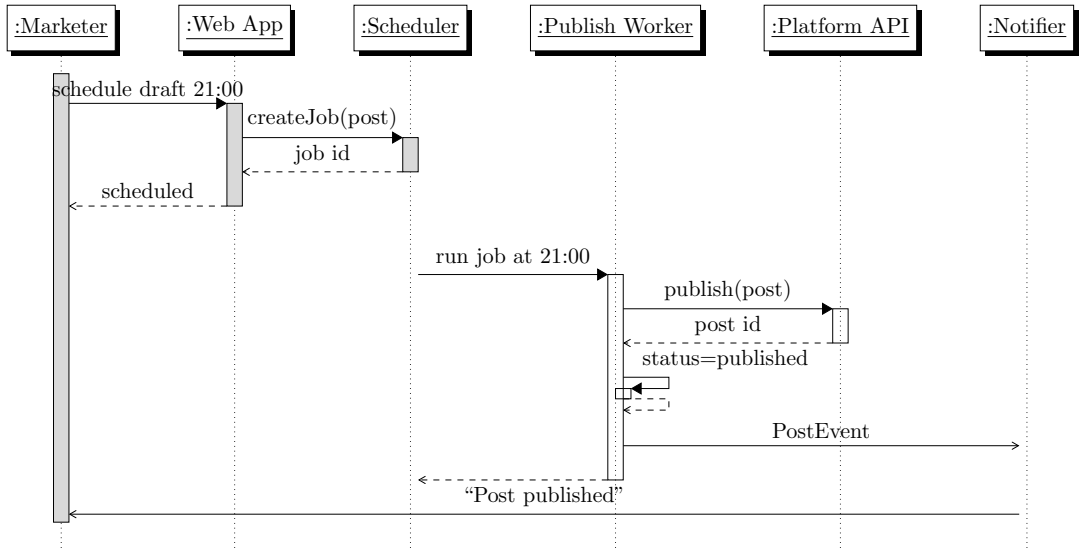


Figure 8: Sequence Diagram for Schedule & Publish Posts

9.5.3 UC-03: View Unified Analytics

In this use case the system collects social and website data and presents it, with Saudi-native insights, in one dashboard.

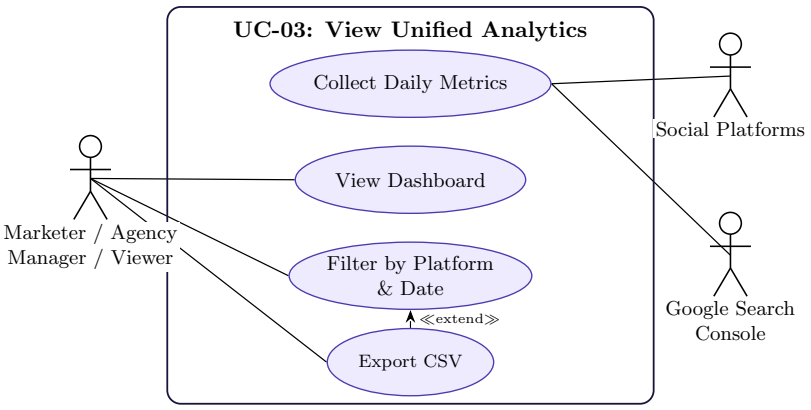


Figure 9: Use Case Diagram: View Unified Analytics

Use Case	**UC-03 View Unified Analytics**
Actors	Marketer, Agency Manager or Viewer (primary); Social Platforms, Google Search Console, Scheduler (secondary)
Preconditions	Accounts and website property are connected; at least one day of data has been collected.
Trigger	User opens the Dashboard or Analytics page.
Main Flow	1. Every day at 06:00 (Saudi time) the Scheduler runs collectors that pull metrics from each platform and Search Console. 2. System stores daily metrics per account and refreshes aggregate views. 3. User opens the dashboard and selects a date range and platforms. 4. System aggregates reach, engagement, followers, clicks, impressions and queries. 5. System computes Saudi-native insights (best times to post, rising queries, upcoming occasions). 6. System displays KPI cards, charts, top posts, search queries and insights in the chosen language within 2 seconds.
Alternative Flows	1a. A source is disconnected or over quota: system shows last sync time and a reconnect banner. 3a. No data yet: system shows an onboarding empty state. 6a. User exports the data as CSV.
Postconditions	User sees one up-to-date view of social and website performance.
Requirements	RQ-F2, RQ-F6, RQ-F7, RQ-F13, RQ-NF1, RQ-NF2, RQ-NF5, RQ-NF6, RQ-NF7

Table 35: Use Case Description: UC-03 View Unified Analytics

Use Case	**UC-03 View Unified Analytics**
Actors	Marketer, Agency Manager or Viewer (primary); Social Platforms, Google Search Console, Scheduler (secondary)
Preconditions	Accounts and website property are connected; at least one day of data has been collected.
Trigger	User opens the Dashboard or Analytics page.
Main Flow	1. Every day the Scheduler runs collectors that pull metrics from each platform and Search Console. 2. System stores daily metrics per account and refreshes aggregate views. 3. User opens the dashboard and selects a date range and platforms. 4. System aggregates reach, engagement, followers, clicks, impressions and queries. 5. System computes Saudi-native insights (best time to post, trending queries, upcoming occasions). 6. System displays KPI cards, charts, top posts, search queries and insights in the chosen language.
Alternative Flows	1a. A source is disconnected or over quota: system shows last sync time and a reconnect banner. 3a. No data yet: system shows an onboarding empty state. 6a. User exports the data as CSV.
Postconditions	User sees one up-to-date view of social and website performance.
Requirements	RQ-F2, RQ-F6, RQ-F7, RQ-F13, RQ-NF1, RQ-NF2, RQ-NF5, RQ-NF6, RQ-NF7

Figure 10: Sequence Diagram for View Unified Analytics

9.5.4 UC-04: Generate Bilingual Report

This use case produces the weekly performance report in Arabic and/or English, optionally branded with the client's logo.

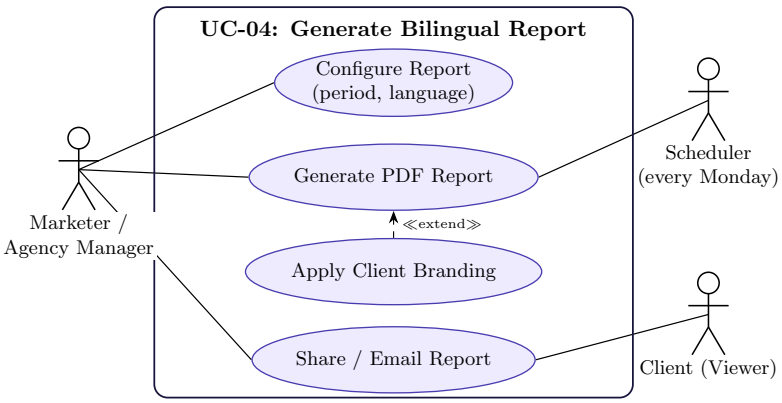


Figure 11: Use Case Diagram: Generate Bilingual Report

Use Case	UC-04 Generate Bilingual Report
Actors	Marketer or Agency Manager (primary); Scheduler, Client/Viewer (secondary)
Preconditions	Workspace has collected metrics for the selected period.
Trigger	User clicks "Generate PDF report", or the Scheduler triggers the Monday 09:00 auto-report.
Main Flow	1. User selects workspace, period, language (Arabic, English or both) and sections. 2. System loads aggregated metrics and top posts for the period. 3. System asks the AI service for a short executive summary in each selected language. 4. System fills the report template (RTL for Arabic) and applies client branding if enabled. 5. System renders the PDF with Arabic-capable fonts and stores it. 6. System notifies the user and shares a link or emails the PDF to the client.
Alternative Flows	1a. Branding requested on Starter plan: system offers the Agency upgrade. 2a. A source has missing days: report marks the gap instead of failing.
Postconditions	A bilingual PDF report is stored in the workspace and delivered.
Requirements	RQ-F8, RQ-F11, RQ-F13, RQ-NF1, RQ-NF6

Table 36: Use Case Description: UC-04 Generate Bilingual Report

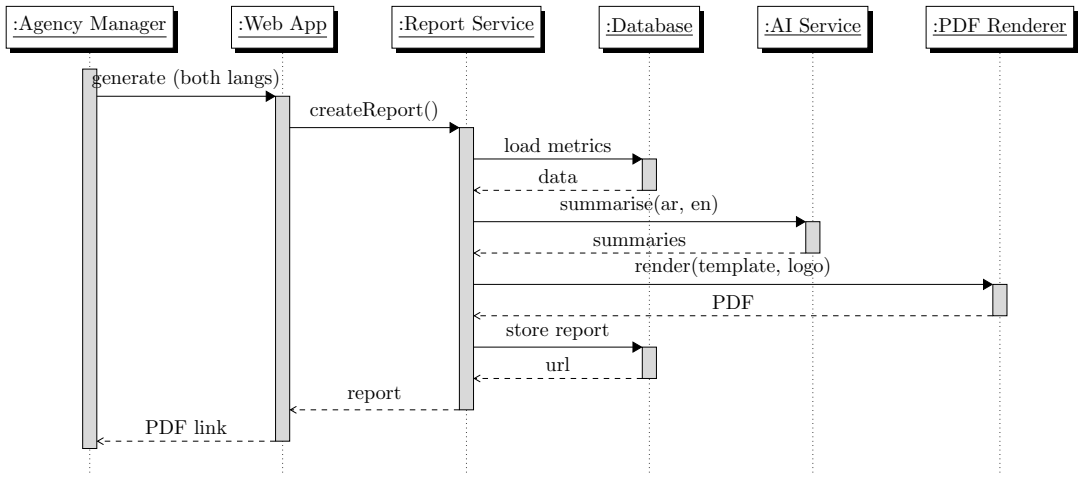


Figure 12: Sequence Diagram for Generate Bilingual Report

9.5.5 UC-05: Manage Client Workspaces & Team Roles

This use case lets agencies separate each client's data and control what each team member can do.

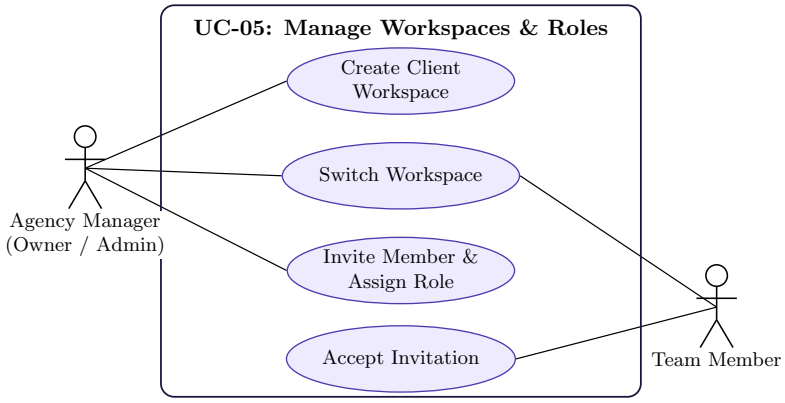


Figure 13: Use Case Diagram: Manage Workspaces & Roles

Use Case	**UC-05 Manage Client Workspaces & Team Roles**
Actors	Agency Manager – Owner/Admin (primary); Team Member (secondary)
Preconditions	User is on the Agency plan and has the Owner or Admin role.
Trigger	User clicks "New client" or "Invite member".
Main Flow	1. Agency Manager creates a client workspace (name, logo, default language). 2. Agency Manager connects the client's social accounts and website (UC-06 / RQ-F2). 3. Agency Manager invites a team member by email and assigns a role (Admin, Editor or Viewer) per workspace. 4. System sends the invitation; the member accepts and sets a password. 5. System applies permissions; the member sees only the allowed workspaces. 6. Users switch between allowed workspaces from the top bar in under 2 seconds without logging out.
Alternative Flows	1a. Workspace limit of the plan reached: system offers an upgrade. 4a. Invitation expires after 7 days: Admin can resend it.
Postconditions	Workspace and memberships are stored; access is enforced on every request.
Requirements	RQ-F9, RQ-F10, RQ-F11, RQ-NF4

Table 37: Use Case Description: UC-05 Manage Client Workspaces & Team Roles

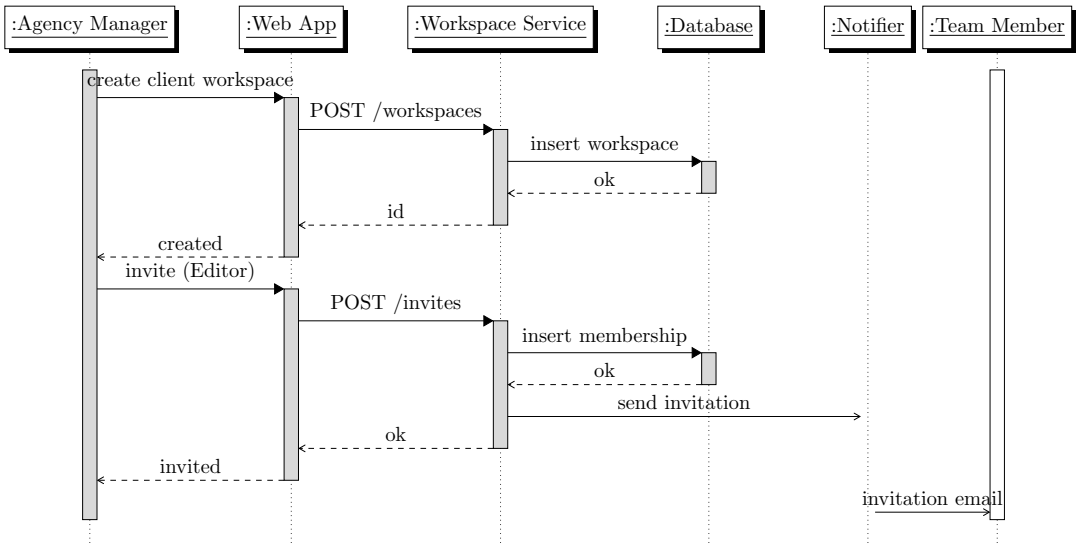


Figure 14: Sequence Diagram for Manage Workspaces & Roles

9.5.6 UC-06: Manage Account, Security & Subscription

This use case covers sign-up, security settings and paying for a plan in SAR.

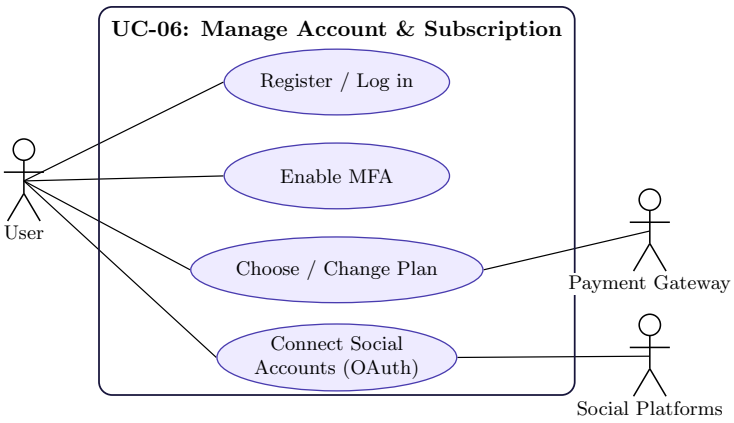


Figure 15: Use Case Diagram: Manage Account & Subscription

Use Case	**UC-06 Manage Account, Security & Subscription**
Actors	User (primary); Payment Gateway, Social Platforms (secondary)
Preconditions	None for registration; logged in for other steps.
Trigger	User signs up, opens Settings, or selects a plan.
Main Flow	1. User registers with name, email, password, user type and interface language. 2. System verifies the email (link valid 24 hours) and creates a personal workspace. 3. User optionally enables MFA (authenticator app). 4. User chooses a plan; system sends the SAR amount plus 15% VAT to the Payment Gateway. 5. Payment Gateway confirms payment; system activates the subscription and issues a VAT invoice within 1 minute. 6. User connects social accounts through OAuth; system stores encrypted tokens with minimum scopes.
Alternative Flows	3a. Lost MFA device: user signs in with a recovery code. 4a. Payment declined: system keeps the Free plan and shows the gateway message. 6a. User revokes a connection: tokens are deleted within 1 minute.
Postconditions	Account, subscription and connections are active and secured.
Requirements	RQ-F1, RQ-F2, RQ-F12, RQ-F13, RQ-NF4, RQ-NF5

Table 38: Use Case Description: UC-06 Manage Account, Security & Subscription

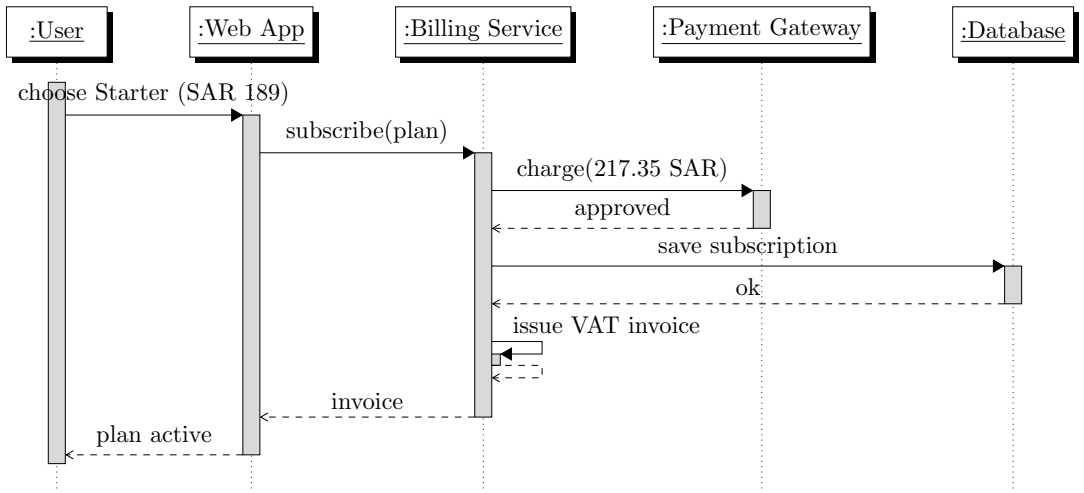


Figure 16: Sequence Diagram for Subscribe to a Plan (SAR 189 + 15% VAT)

10 Lab 4: Software Requirements Specification (SRS)

10.1 Introduction

This document specifies the software requirements for Tempo, an AI-powered bilingual (Arabic/English) marketing co-pilot for solo marketers, SMEs, and digital agencies in Saudi Arabia. Tempo integrates four platforms — Snapchat, X, Instagram, and Google Search Console — into a single bilingual dashboard. This document consolidates the use cases identified in Lab 3 into four simplified, high-priority scenarios and derives the functional and non-functional requirements that the system must satisfy. All requirements are traceable to the user personas (Nora Al-Shamrani and Khalid Al-Otaibi) and the use cases defined here. For the midterm, every requirement in this SRS was revised to be SMART and aligned with Section 8 (see Table 30).

10.2 Lab Objectives

1. Simplify and consolidate use cases into four core workflows.
2. Define functional requirements (FR) organised by system module.
3. Specify non-functional requirements (NFR) covering performance, security, and usability.
4. Produce a requirements traceability matrix linking personas → use cases → requirements.

10.3 Part 1: Simplified Use Cases

The six use cases from Lab 3 (Section 9) have been consolidated into four core use cases covering the primary workflows of Tempo. Each is presented in brief format (name, actor, trigger, main flow, outcome).

10.3.1 Use Case Diagram

Figure 17 shows all four simplified use cases, the primary actors (Nora and Khalid), and the external system actors they interact with.

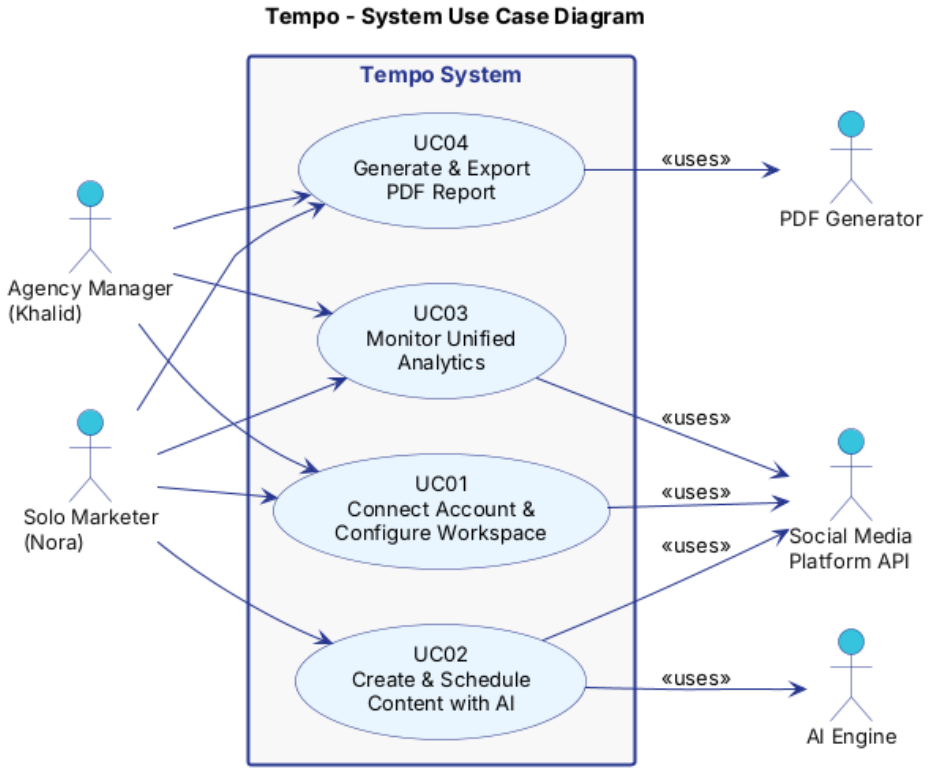


Figure 17: Lab 4 – Tempo System Use Case Diagram

10.3.2 Use Case Overview

UC	Name	Primary Actor	Trigger	Outcome
L4-UC01	Connect Account & Configure Workspace	Nora / Khalid	User signs up or logs in	Social accounts linked; workspace ready
L4-UC02	Create & Schedule Content with AI	Nora	User clicks "Generate Content"	Post scheduled on selected platforms
L4-UC03	Monitor Unified Analytics	Nora / Khalid	User opens Dashboard	Cross-platform metrics displayed; alerts triggered
L4-UC04	Generate & Export PDF Report	Khalid / Nora	User clicks "Generate Report"	Branded bilingual PDF downloaded or shared

Table 39: Lab 4 Use Case Overview

Field	Description
Use Case ID	L4-UC01
Use Case Name	Connect Account & Configure Workspace
Actors	Solo Marketer (Nora), Agency Manager (Khalid)
Precondition	User has a Tempo account (free or paid) and at least one social platform account.
Main Flow	1. User navigates to Settings → Connected Accounts. 2. Selects a platform (Instagram, X, Snapchat, or Google Search Console). 3. Authenticates via OAuth 2.0 and grants required permissions. 4. Tempo confirms the connection and displays the account handle / property within 10 seconds. 5. Agency user additionally creates or names a client workspace and assigns team members.
Alternative Flow	OAuth fails → error message shown; user retries or contacts support.
Postcondition	Social account linked to workspace; first data sync starts within 5 minutes.
Priority	Must Have

Table 40: L4-UC01 – Connect Account & Configure Workspace

L4-UC01 – Connect Account & Configure Workspace

Field	Description
Use Case ID	L4-UC02
Use Case Name	Create & Schedule Content with AI
Actors	Solo Marketer (Nora), AI Engine, Social Media Platform API
Precondition	At least one social account is connected; user has remaining AI generation quota.
Main Flow	1. User selects platform and language (Arabic / English / Bilingual). 2. Clicks "Generate Content"; optionally adds a brief or theme. 3. AI Engine analyses the top 10 posts of the last 30 days and returns 3 content variants with engagement scores (0–100) within 30 seconds. 4. User reviews variants, optionally edits, then selects one. 5. User sets publish date/time and clicks "Schedule". 6. Tempo queues the post and confirms the scheduled time.
Alternative Flow	Fewer than 5 past posts → AI uses brand style guide template. Platform API error → post saved as draft with error notification.
Postcondition	Post queued in content calendar; platform notified at scheduled time.
Priority	Must Have

Table 41: L4-UC02 – Create & Schedule Content with AI

L4-UC02 – Create & Schedule Content with AI

Activity Diagram – L4-UC02 Figure 18 shows the detailed activity flow for L4-UC02, including the parallel AI processing steps and the decision branches for user satisfaction and publish success.

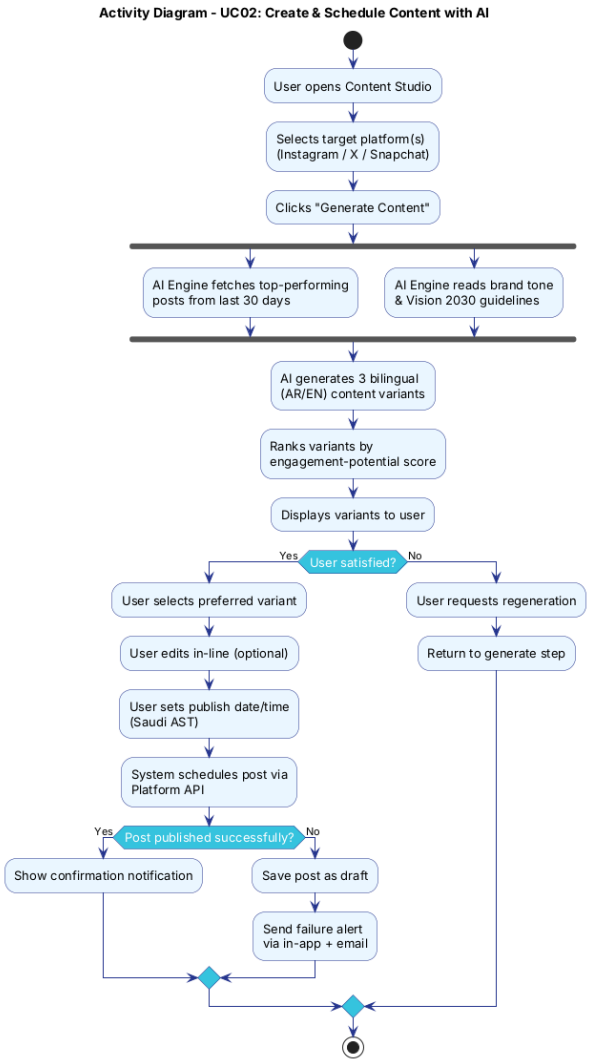


Figure 18: Activity Diagram: Create & Schedule Content with AI (L4-UC02)

Field	Description
Use Case ID	L4-UC03
Use Case Name	Monitor Unified Analytics
Actors	Solo Marketer (Nora Al-Shamrani), Agency Manager (Khalid Al-Otaibi), Analytics Service
Precondition	At least one social account and/or Google Search Console property connected; at least 24 hours of data collected.
Main Flow	1. User opens the Dashboard tab. 2. System displays cross-platform social metrics (reach, engagement rate, follower growth) alongside Google Search Console website data (clicks, impressions, CTR). 3. User selects a date range (7 days / 30 days / custom). 4. Dashboard updates the charts within 2 seconds. 5. If any metric drops more than 20% week-over-week, a performance alert is displayed.
Alternative Flow	No data for selected range → placeholder message; system prompts to connect account or wait for sync.
Postcondition	User has an up-to-date cross-platform performance overview; alerts acknowledged or dismissed.
Priority	Must Have

Table 42: L4-UC03 – Monitor Unified Analytics

L4-UC03 – Monitor Unified Analytics

Field	Description
Use Case ID	L4-UC04
Use Case Name	Generate & Export PDF Report
Actors	Agency Manager (Khalid), Solo Marketer (Nora)
Precondition	Analytics data available for the selected period; branding assets configured for agency workspace.
Main Flow	1. User clicks “Generate Report” and selects the reporting period. 2. Chooses language (Arabic / English / Both) and report type (Summary / Full). 3. System compiles reach, engagement, top posts, and platform breakdown. 4. PDF is rendered within 60 seconds with client branding and correct RTL/LTR layout. 5. User downloads the PDF or copies a share link (valid 7 days).
Alternative Flow	No data for period → report generated with a “No data available” section and a note to the user.
Postcondition	Branded bilingual PDF saved and/or shared; report log updated.
Priority	Must Have

Table 43: L4-UC04 – Generate & Export PDF Report

L4-UC04 – Generate & Export PDF Report

10.4 Part 2: Functional Requirements

Functional requirements are organised by system module. Each requirement is identified by a unique ID and linked to the use case it supports. Rows shaded green were rewritten in the SMART revision.

10.4.1 Module A – Authentication & Workspace Management

FR ID	Requirement	Linked UC	Priority
L4-FR-A01	The system shall allow users to register and log in using email/password with optional multi-factor authentication (authenticator app), and shall lock an account for 15 minutes after 5 failed login attempts.	L4-UC01	Must Have
L4-FR-A02	The system shall allow users to connect social accounts (Instagram, X, Snapchat) and Google Search Console properties via OAuth 2.0, confirm the connection within 10 seconds, and start the first data sync within 5 minutes.	L4-UC01	Must Have
L4-FR-A03	Agency users shall be able to create at least 10 client workspaces and switch between them in under 2 seconds without logging out.	L4-UC01	Must Have
L4-FR-A04	The system shall support role-based access control with roles Owner, Admin, Editor, and Viewer; 100% of actions outside a user's role shall be refused, verified by automated tests.	L4-UC01	Must Have
L4-FR-A05	Users shall be able to revoke social account connections at any time from the Settings panel; the stored tokens shall be deleted within 1 minute.	L4-UC01	Should Have

Table 44: Module A – Authentication & Workspace Management

10.4.2 Module B – AI Content Generation & Scheduling

FR ID	Requirement	Linked UC	Priority
L4-FR-B01	The system shall generate exactly 3 content variants per request within 30 seconds using a commercial LLM API, in Arabic, English, or both.	L4-UC02	Must Have
L4-FR-B02	Content generation shall use the user's 10 posts with the highest engagement rate in the past 30 days as context for the prompt (or the brand style template if there are fewer than 5 posts).	L4-UC02	Must Have
L4-FR-B03	Generated content shall include an engagement-potential score from 0 to 100 for each variant, ranked highest first.	L4-UC02	Should Have
L4-FR-B04	Users shall be able to edit any generated variant in-line before scheduling, with a live character count against each platform's limit.	L4-UC02	Must Have
L4-FR-B05	The system shall schedule posts to Instagram, X, and Snapchat at a user-specified date and time (Saudi AST), to the minute, up to 90 days ahead.	L4-UC02	Must Have
L4-FR-B06	If a scheduled post fails to publish after 3 retries, the system shall notify the user by in-app alert and email within 2 minutes, and save the post as a draft.	L4-UC02	Must Have

Table 45: Module B – AI Content Generation & Scheduling

10.4.3 Module C – Unified Analytics Dashboard

FR ID	Requirement	Linked UC	Priority
L4-FR-C01	The system shall display a unified dashboard showing reach, engagement rate, and follower growth for all connected social platforms (Snapchat, X, Instagram).	L4-UC03	Must Have
L4-FR-C02	The dashboard shall support date-range filtering: last 7 days, last 30 days, and a custom range of up to 12 months.	L4-UC03	Must Have
L4-FR-C03	The system shall collect platform metrics automatically via scheduled API calls once every 24 hours, completed by 06:00 AST.	L4-UC03	Must Have
L4-FR-C04	The system shall display a "Top Performing Post" card per platform, updated daily by 07:00 AST.	L4-UC03	Should Have
L4-FR-C05	The system shall trigger a performance alert within 1 hour of the daily sync when any metric drops more than 20% week-over-week.	L4-UC03	Should Have
L4-FR-C06	The system shall display Google Search Console data (queries, clicks, impressions, and average position) alongside social metrics in the unified dashboard.	L4-UC03	Must Have

Table 46: Module C – Unified Analytics Dashboard

10.4.4 Module D – PDF Report Generation

FR ID	Requirement	Linked UC	Priority
L4-FR-D01	The system shall generate a PDF performance report for any selected date range of up to 12 months within 60 seconds of the user's request.	L4-UC04	Must Have
L4-FR-D02	Reports shall be available in Arabic (RTL layout), English (LTR layout), or both languages in a single document.	L4-UC04	Must Have
L4-FR-D03	Agency workspace reports shall include the client logo and two brand colours configured by the agency.	L4-UC04	Must Have
L4-FR-D04	Reports shall include: Executive Summary, Platform Breakdown, Top 5 Posts, and at least 3 Recommendations.	L4-UC04	Should Have
L4-FR-D05	The system shall allow reports to be downloaded as PDF or shared via a secure link valid for 7 days.	L4-UC04	Should Have

Table 47: Module D – PDF Report Generation

10.5 Part 3: Non-Functional Requirements

Non-functional requirements define the quality attributes the system must exhibit regardless of specific functionality. They are organised by category.

NFR ID	Category	Requirement	Priority
L4-NFR-01	Performance	The dashboard shall load and display all platform metrics within 2 seconds (95th percentile) on a 4G or broadband connection.	Must Have
L4-NFR-02	Performance	AI content generation shall return results within 30 seconds (95th percentile) for a standard request (3 variants, 1 platform).	Must Have

NFR ID	Category	Requirement	Priority
L4-NFR-03	Availability	The system shall maintain 99.5% monthly uptime, with no planned maintenance during Saudi peak hours (6 PM – 12 AM AST).	Must Have
L4-NFR-04	Security	All social account OAuth tokens shall be encrypted at rest using AES-256 and transmitted only over HTTPS (TLS 1.2+).	Must Have
L4-NFR-05	Security	The system shall undergo an external penetration test before the public launch and remediate all Critical and High findings.	Must Have
L4-NFR-06	Privacy	The system shall comply with the Saudi PDPL: user data stored in a Saudi-region cloud, deletable on request within 30 days.	Must Have
L4-NFR-07	Usability	The interface shall support right-to-left (RTL) Arabic text alongside left-to-right (LTR) English on 100% of screens, checked against a UI checklist before every release.	Must Have
L4-NFR-08	Usability	At least 80% of closed-beta testers shall connect a social account and generate their first content suggestion within 10 minutes of sign-up.	Should Have
L4-NFR-09	Localisation	All date, time, currency, and calendar displays shall default to Saudi Arabia conventions (AST, SAR, Hijri/Gregorian toggle).	Should Have
L4-NFR-10	Scalability	The architecture shall support horizontal scaling to serve at least 1,000 concurrent users without degrading below L4-NFR-01/L4-NFR-02 thresholds, verified by load test before January 2028.	Should Have
L4-NFR-11	Maintainability	The AI content module shall be provider-independent, allowing the LLM provider to be swapped in at most 5 working days without changes to other modules.	Should Have

Table 48: Lab 4 Non-Functional Requirements

10.6 Part 4: Requirements Traceability Matrix

The matrix below links each persona to the use case that represents their primary goal, and maps the functional requirements that implement it.

Persona	Use Case	Functional Requirements	NFRs
Nora (Solo Marketer)	L4-UC01: Connect Account	L4-FR-A01, L4-FR-A02, L4-FR-A05	L4-NFR-04, L4-NFR-07
Nora	L4-UC02: Create & Schedule Content	L4-FR-B01, L4-FR-B02, L4-FR-B03, L4-FR-B04, L4-FR-B05, L4-FR-B06	L4-NFR-01, L4-NFR-02, L4-NFR-07, L4-NFR-08
Nora / Khalid	L4-UC03: Monitor Analytics	L4-FR-C01, L4-FR-C02, L4-FR-C03, L4-FR-C04, L4-FR-C05, L4-FR-C06	L4-NFR-01, L4-NFR-03, L4-NFR-07
Khalid (Agency Manager)	L4-UC01: Configure Workspace	L4-FR-A03, L4-FR-A04, L4-FR-A05	L4-NFR-04, L4-NFR-06, L4-NFR-10
Khalid	L4-UC04: Generate PDF Report	L4-FR-D01, L4-FR-D02, L4-FR-D03, L4-FR-D04, L4-FR-D05	L4-NFR-06, L4-NFR-07, L4-NFR-09

Table 49: Lab 4 Requirements Traceability Matrix

11 Context Diagram

The context diagram (Level 0 data flow diagram) shows Tempo as a single process and the data it exchanges with its three user types and four external systems. It defines the boundary of the system: everything inside the circle is built by the team; everything outside is a user or an external service.

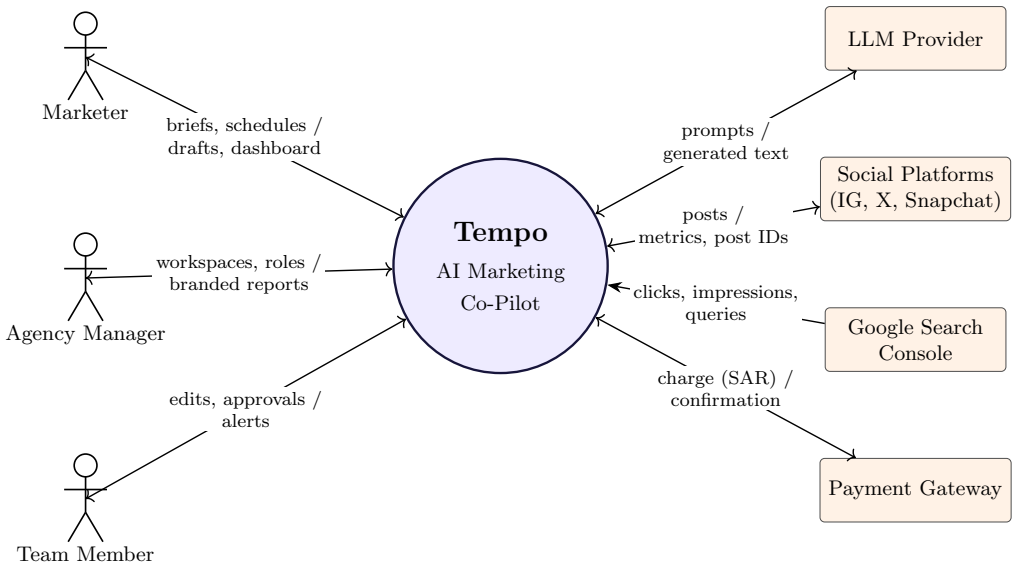


Figure 19: System Context Diagram (Level 0)

External entity	Data into Tempo	Data out of Tempo
Marketer	Briefs, schedules	Drafts, dashboard
Agency Manager	Workspaces, roles	Branded reports
Team Member	Edits, approvals	Alerts
LLM Provider	Generated text	Prompts
Social Platforms (IG, X, Snapchat)	Metrics, post IDs	Posts
Google Search Console	Clicks, impressions, queries	— (read-only access)
Payment Gateway	Payment confirmation	Charge request (SAR)

Table 50: Context Diagram Data Flows

12 Project Management

12.1 Methodology

Tempo is developed with **Agile Scrum** in two-week sprints. The Product/Project Manager acts as Product Owner, the product backlog is built from the user stories in Section 9 and prioritised with MoSCoW (Must / Should Have), and each sprint ends with a demo and a retrospective. Platform API access is requested in the first weeks because approvals can take time. If Snapchat approval or the beta runs late, the launch *scope* is reduced (X and Instagram first) rather than the *date* moved.

Scrum element	How Tempo uses it
Product Owner	Product / Project Manager (founder) owns and orders the backlog
Development team	Backend & AI-integration developer and Frontend & UX developer (founders); roles in Table 8
Sprint	Two weeks; Sprint 0 (October 2026) sets up architecture, CI/CD and API applications
Events	Sprint planning, short daily stand-up, sprint review/demo and retrospective
Artefacts	Product backlog (user stories), sprint backlog, burndown chart on the Scrum board
Definition of Done	Code reviewed, unit tests passing (70% coverage on core modules), works in Arabic RTL and English LTR, requirement ID linked
Tools	Jira or GitHub Projects (Scrum board), GitHub Actions (CI/CD), Figma (design), Google Workspace (documents)

Table 51: Scrum Set-up

12.2 Schedule

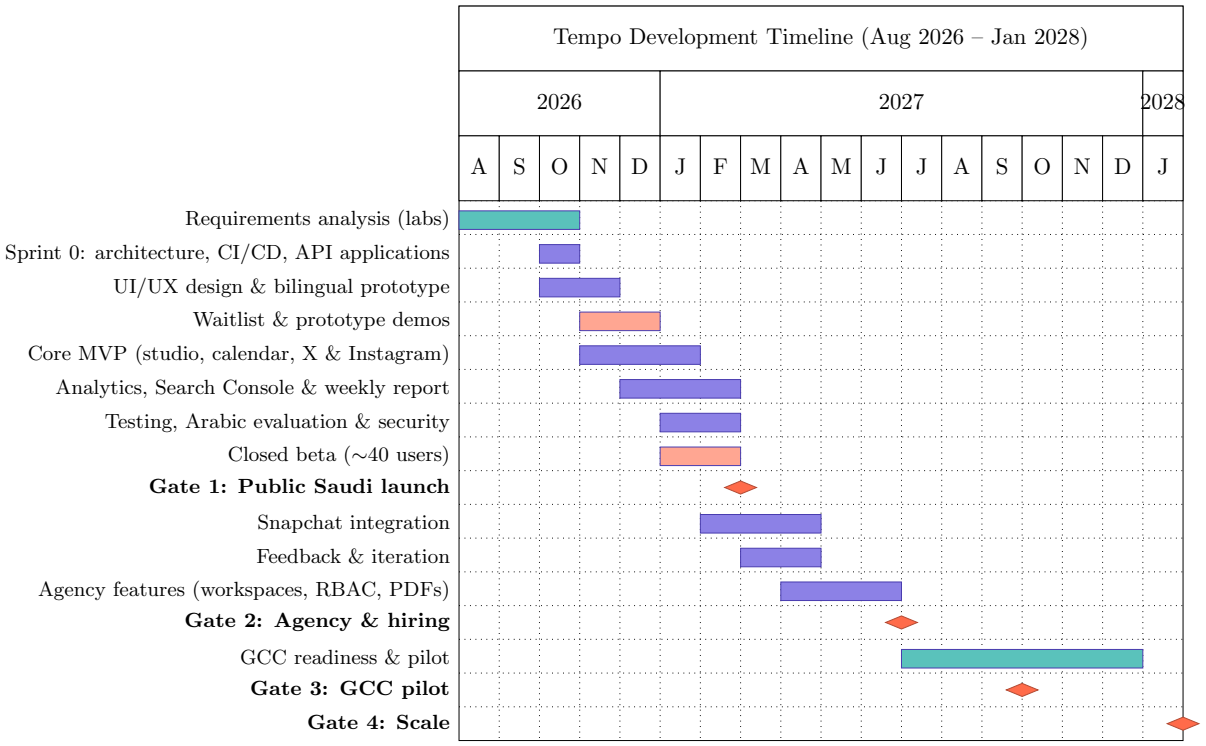


Figure 20: Project Gantt Chart

No.	Milestone	Target date
1	Requirements analysis: market study, personas, feasibility study, use cases, SRS (Requirements Engineering Phase 1)	Aug – Oct 2026
2	Sprint 0: architecture, repositories, CI/CD, cloud; apply for Meta, X, Snap and Google API access	Oct 2026
3	UI/UX design and clickable bilingual prototype	Oct – Nov 2026
4	Waitlist and prototype demos in Riyadh and Jeddah	Nov – Dec 2026
5	Core MVP: sign-up, workspaces, AI Content Studio, calendar, X and Instagram publishing	Nov 2026 – Jan 2027
6	Analytics: Google Search Console, unified dashboard, weekly bilingual report	Dec 2026 – Feb 2027
7	Testing: automated tests, Arabic content evaluation, security review, penetration test	Jan – Feb 2027
8	Closed beta with about 40 invited users	Jan – Feb 2027
9	Public Saudi launch: freemium and Starter (Riyadh, Jeddah, Dammam)	End of Feb 2027
10	Snapchat integration (subject to platform approval)	Feb – Apr 2027
11	Feedback and iteration: monthly releases	Mar – Apr 2027
12	Agency features: client workspaces, RBAC, branded bilingual PDF reports	Apr – Jun 2027
13	GCC readiness and pilot, if validation gate is met	Jul – Dec 2027

Table 52: Implementation Milestones

12.3 Risk and Budget Control

Project risks are tracked in the risk register (Table 19) and reviewed at every sprint retrospective; each risk has an owner and a mitigation. Spending is released only when a stage gate is met (Table 21): public launch (February 2027), agency features and hiring (June 2027), GCC pilot (September 2027) and scale (January 2028). This limits the money at risk to about SAR 115,000 before launch.

12.4 Requirements Change Control

The requirements in this report are the **baseline** for Phase 1. After the midterm, any change follows four steps: (1) the change is logged with an ID, the reason and who asked for it; (2) the team checks its impact on use cases, schedule and cost using the traceability matrices (Tables 32 and 49); (3) the Product Owner accepts or rejects it at sprint planning; (4) the accepted change is rewritten in SMART form and the matrices are updated.

13 Conclusion

Tempo provides a focused solution to a clear problem: Saudi marketers lose hours every week rewriting English-first AI output, switching between four platforms and building reports by hand. By combining Saudi-dialect AI content, scheduling for Snapchat, X and Instagram, unified social and website analytics and automated bilingual reports at an SME price in SAR, Tempo fills a gap no incumbent covers. The analysis phase has delivered:

- A clear problem statement and strong market potential with competitive differentiation (SAR 3.5B TAM, 68% demand for all-in-one dashboards)
- A feasibility study showing technical and operational feasibility, and conditional economic feasibility with break-even at about 614 paying accounts in mid-2029, controlled by stage gates
- A business model canvas linking segments, value, channels, costs and revenue
- User-centred requirements derived from three personas and eight elicitation methods
- 13 functional and 8 non-functional requirements, plus the Lab 4 SRS, all revised to be SMART and consistent with each other
- Six use cases with diagrams, user stories and traceability, a context diagram, and a Scrum-based project management plan

The next phase moves to system design (architecture, data model and user interface) and to validating demand and Arabic output quality with the prototype and the closed beta.

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